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Comparison and Implementation of Data Compression Methods for SoC-based Radar Systems

by

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Abstract

High-resolution cascaded MIMO radar produces data faster than a vehicle’s network and central processor can comfortably absorb it. Compressing that data at the sensor, before it enters the network, is an attractive way to buy back bandwidth — but only if the compressor is fast, cheap in silicon, and lossless, because a radar that quietly discards weak targets to save bits is not a safe radar.

This thesis implements, verifies and critically evaluates two lossless compressors for radar range-FFT data on an AMD Kintex UltraScale+ FPGA: *Doppler Redundancy Hybrid Encoding* (DRHE), a streaming polar-domain predictor, and a second-order *Linear Predictive Coder* (LPC) with Yule-Walker coefficients. Both share a category-plus-append entropy back end with a fixed Huffman dictionary. Both were developed in MATLAB against a double-precision reference, translated to C++ for Vitis HLS, verified bit-exactly over 50 frames of the real ColoRadar cascade dataset, and characterised through RTL co-simulation.

The headline results are not the ones that were expected. A systematic audit of the compression ratio found that *neither predictor was contributing anything*: coding the quantised FFT output through the identical entropy coder with no prediction whatsoever achieves a compression ratio of 3.489, against 3.319 for DRHE and 3.397 for LPC. Both predictors were making the data measurably harder to compress. The cause was traced to an assumption inherited from the single-transmitter literature. In a twelve-transmitter time-division MIMO cascade the ramp index interleaves transmitters, so “the previous ramp” belongs to a different physical antenna eleven times out of twelve, separated in space by the array baseline rather than in time by a pulse interval. Predicting instead from the previous ramp of the *same* transmitter — lag n_{Tx} rather than lag 1 — converts a 4.9% loss into an 8.2% gain. Measured in hardware simulation over the full dataset, compression rises from 3.310 to 3.751 for DRHE and from 3.385 to 3.697 for LPC, in both cases with every sample still reconstructed exactly. For DRHE the correction costs 40 additional block RAMs — 4% of the device — with logic, clock frequency and measured frame latency all unchanged.

Two implementation defects were also found and fixed, neither visible on real data and both caught by synthetic edge cases: a residual value of -32768 that the entropy code cannot represent and which aliased silently onto $+32767$, and a 128-bit to single-precision conversion in the Yule-Walker solver that returned zero for operands above 2^{64} . A throughput figure widely quoted for this class of design was also found to be expressed in gibibits rather than gigabits, understating the performance gap by 7.4%.

The thesis closes with a decomposition of where the compression ratio genuinely comes from — $1.63\times$ from an under-filled fixed-point container, $2.14\times$ from spectral sparsity, and only then a modest contribution from prediction — and argues that reporting a no-prediction baseline alongside any such ratio should be standard practice.

Acknowledgements

I would like to thank A/Prof. Beena Ahmed for her guidance throughout this project, and Dr. Mostafa Shahin for his assessment and feedback. I am also indebted to Lukas Kiem, whose master's thesis provided the algorithmic baseline this work builds on and corrects; the fact that this thesis identifies a limitation in applying that work to MIMO radar should be read as a measure of how carefully specified it was, since a vaguer description could not have been tested against measurement at all.

Abbreviations

ADAS	Advanced Driver Assistance Systems
ADC	Analog-to-Digital Converter
AR	Autoregressive
ASIC	Application-Specific Integrated Circuit
AXI	Advanced eXtensible Interface
BAQ	Block Adaptive Quantization
BFP	Block Floating Point
BRAM	Block Random Access Memory
CFAR	Constant False Alarm Rate
CR	Compression Ratio
CSIM	C-Simulation
DDR	Double Data Rate (memory)
DPCM	Differential Pulse-Code Modulation
DRHE	Doppler Redundancy Hybrid Encoding
DSP	Digital Signal Processing / DSP slice
ECU	Electronic Control Unit
EGE	Exponential-Golomb Encoding
EVM	Evaluation Module
FDR	False Discovery Rate
FF	Flip-Flop
FFT	Fast Fourier Transform
FMCW	Frequency-Modulated Continuous Wave
FN	False Negative
FNR	False Negative Rate
FP	False Positive
FPGA	Field-Programmable Gate Array
FPU	Floating-Point Unit
FX16	16-bit Fixed-point representation
HDL	Hardware Description Language
HLS	High-Level Synthesis
IF	Intermediate Frequency
IIR	Infinite Impulse Response
II	Initiation Interval
I/Q	In-phase / Quadrature
LPC	Linear Predictive Coding
LSB	Least Significant Bit
LUT	Look-Up Table
MIMO	Multiple-Input Multiple-Output
MSB	Most Significant Bit
PRI	Pulse Repetition Interval
RTL	Register-Transfer Level

RX	Receive / Receiver
SAR	Synthetic Aperture Radar
SerDes	Serialiser/Deserialiser
SNR	Signal-to-Noise Ratio
SoC	System-on-Chip
TDM	Time-Division Multiplexing
TX	Transmit / Transmitter
URAM	UltraRAM

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Chapter 1

Introduction

1.1 The problem in one paragraph

A modern high-resolution automotive radar generates data faster than the vehicle it is bolted to can conveniently move it. The sensor at the centre of this thesis produces 3 MiB of raw digitised samples every frame, and it is one of several such sensors on a car. Getting that data to the central processor means either expensive shielded high-speed cabling, or compressing it at the sensor first. Compression is the cheaper answer, but it has to be done in a few hundred microseconds, in a small amount of silicon, and — because this data feeds a system that decides whether to brake — without losing anything. This thesis builds two such compressors, puts them on an FPGA, measures them properly, and finds that the accepted way of applying them to a MIMO radar is wrong.

1.2 Motivation

Radar earns its place in a vehicle’s sensor suite because it works when the other sensors do not. Cameras fail in fog, glare and darkness; lidar is degraded by rain and airborne dust; radar sees through all of them [1, 2]. That robustness is why radar has been in production vehicles for two decades, initially for simple distance warning and now for adaptive cruise control, automatic emergency braking, blind-spot detection and lane-change assistance [3].

What has changed recently is resolution. Supporting SAE Level 2+ autonomy needs more than “something is 40 m ahead”. It needs a map: how many objects, where, how fast, how wide, and what they probably are. Angular resolution is what separates “an object” from “a car and a pedestrian standing beside it”, and angular resolution comes from having many antennas. A radar with N_t transmitters and N_r receivers can synthesise an effective array of $N_t \times N_r$ elements, so resolution improves roughly with the product. The data volume, unfortunately, scales the same way.

At the same time, vehicle electrical architectures have been consolidating. Where a car once had a hundred small electronic control units each doing one job, the industry has moved to domain controllers, then to *zonal* architectures where sensors report to a nearby gateway, and toward fully centralised designs where one high-performance system-on-chip does everything [4, 5, 6]. Centralising the computation has obvious advantages, but it means the raw sensor data has to travel.

Figure 1.1 shows the consequence. A Serialiser/Deserialiser (SerDes) link [7, 8] carries the aggregated output of several radar modules at 2.5 Gbit/s or more. Those data rates are achievable, but they require shielded cable, they cost money and weight, and they put a matching ingest burden on the SoC’s memory system [9]. If instead the data is compressed at the radar module by a factor of 2.5 or better, the same information fits inside Gigabit

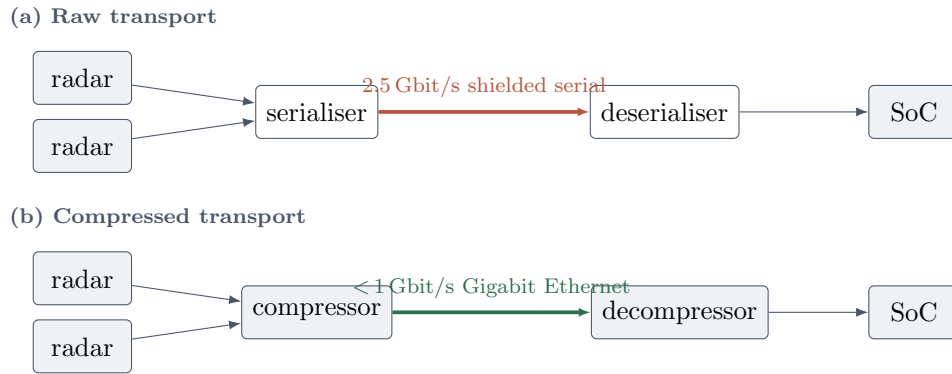


Figure 1.1: The economic argument for sensor-edge compression. Compressing by a factor of 2.5 or better at the radar module replaces an expensive shielded high-speed serial link with standard Gigabit Ethernet, without changing what the central processor receives.

Ethernet [10] — a cheap, standard, unshielded interconnect. That single substitution is the economic argument for this entire body of work.

1.3 Why the compressor must be lossless

It is tempting to relax the requirement. Lossy compression reaches much higher ratios, and radar data is noisy anyway, so surely a little extra noise is harmless?

It is not, and the reason is specific rather than philosophical. The compressor sits *upstream* of detection. Whatever it discards is discarded before the detector ever sees it, and the detector’s job is to find weak returns close to the noise floor — a pedestrian at range, a motorcycle between two trucks. Lossy compression raises the noise floor precisely where the margin is thinnest. Chapter 6 quantifies this: Block Adaptive Quantization, a respected lossy scheme, reaches a compression ratio comparable to the lossless coders studied here but misses nearly three times as many real detections and invents nearly nine times as many false ones.

Losslessness also has a practical virtue that matters enormously for a project of this kind: it makes verification unambiguous. The pass criterion is that every reconstructed sample equals its original exactly. There is no tolerance to argue about, no distortion metric to trade off. A single differing bit anywhere in 157 million samples is a defect, and the test reports it as one. Section 5.4 describes two defects that were found exactly this way and would have been invisible under any tolerance-based criterion.

1.4 Objectives

The starting point for this thesis is the master’s work of Lukas Kiem [11], which evaluated a wide range of radar compression schemes and identified Doppler Redundancy Hybrid Encoding as the best of them, reaching a compression ratio of 3.2 at a latency of 0.08 ms on a Zynq UltraScale+ device. Kiem’s work left three openings, which became the objectives of this thesis:

1. **Validate on real data.** Kiem’s FPGA implementation was exercised with synthetic stimulus. This thesis drives both compressors with real captured data — 50 consecutive frames from a Texas Instruments MMWCAS-RF-EVM cascade, taken from the public ColoRadar dataset [12] — through the entire verification chain, from MATLAB to RTL co-simulation.
2. **Broaden the algorithm comparison.** DRHE is compared here not only against the schemes Kiem considered but against a second-order linear predictive coder with

Yule–Walker coefficients, and against a *no-prediction* baseline that turns out to be the most informative comparison of all.

3. **Implement the complete pipeline.** Kiem implemented compression only. This thesis implements compression *and* decompression for both algorithms, which is what makes bit-exact verification possible at every level, and which exposed a subtle encoder/decoder divergence described in Section 5.4.1.

1.5 Contributions

The substantive contributions of this thesis are:

1. **A working, bit-exact, fully verified FPGA implementation of two lossless radar compressors**, each with a matching decompressor, verified over 50 frames of real cascade radar data at three levels of abstraction (MATLAB reference, HLS C-simulation, RTL co-simulation) with zero reconstruction error at every level.
2. **The identification and correction of a prediction-axis error** in applying slow-time predictive compression to time-division MIMO radar. Both algorithms, applied the way the literature describes, performed *worse than no prediction at all*. Correcting the prediction lag to the transmitter period recovers the intended gain, lifting measured compression from 3.310 to 3.751 for DRHE and 3.385 to 3.697 for LPC. This is, to the author’s knowledge, not reported elsewhere.
3. **A decomposition of the compression ratio** into its three genuine sources — unused fixed-point container headroom, spectral sparsity, and prediction — which shows that the headline ratios reported in this field are substantially a property of sensor gain staging rather than of the algorithm.
4. **Two implementation defects found and fixed**, both invisible on real data and both caught by synthetic edge-case testing: an unrepresentable residual value that aliased onto a different value, and a wide-integer to floating-point conversion that silently returned zero.
5. **Three corrections to the surrounding literature and to the author’s own earlier reporting**: a throughput figure quoted in gibibits and compared against decimal figures; a detection metric labelled as a false positive rate that is in fact a false discovery rate; and two malformed source citations.

1.6 How to read this report

The report is written to be self-contained for a reader who knows some signal processing and digital design but nothing in particular about radar or about compression.

Chapter 2 builds up the necessary background from first principles: how an FMCW radar turns a frequency sweep into a range measurement, what “slow time” and Doppler mean, how MIMO and time-division multiplexing produce a virtual array, what is actually in the data that has to be compressed, how lossless compression works in general, and what an FPGA gives you and charges you for. A reader already fluent in these can skip to Section 2.8.

Chapter 3 specifies the two algorithms and the entropy coder they share, in enough detail to reimplement them.

Chapter 4 sets out the four-phase verification framework and defines every metric used, including several that are commonly used loosely.

Chapter 5 describes the hardware architectures and the two defects.

Chapter 6 reports results, and is where the audit and the prediction-axis correction appear.

Chapter 7 discusses what the numbers support and what they do not, and **Chapter 8** concludes.

Chapter 2

Background

This chapter assumes no prior knowledge of radar. It builds up, in order, what the radar measures, how it measures it, what the resulting data looks like, why there is so much of it, how lossless compression works in general, and what an FPGA can do about it. Each section ends with the specific fact that the later chapters depend on.

2.1 How an FMCW radar measures range

2.1.1 The chirp

Automotive radar is almost universally *frequency-modulated continuous wave* (FMCW). Instead of emitting a short pulse and timing the echo — which would demand impractically fast electronics for the distances involved — an FMCW radar transmits continuously, sweeping its frequency linearly upward over a short interval. One such sweep is called a *chirp*. The sensor used here sweeps 1.798 GHz of bandwidth around a 77 GHz centre frequency in 78.5 μ s.

When the chirp reflects off an object at range R , the echo arrives delayed by the round-trip time

$$\tau = \frac{2R}{c}, \quad (2.1)$$

where c is the speed of light. Because the transmitter has kept sweeping in the meantime, the echo's frequency at any instant is lower than what is currently being transmitted, by an amount proportional to the delay. If the chirp sweeps B hertz in T_c seconds, the sweep slope is $S = B/T_c$ and the frequency difference is

$$f_b = S\tau = \frac{2SR}{c}. \quad (2.2)$$

A mixer multiplies the transmitted and received signals, producing a low-frequency *beat* or intermediate-frequency signal at exactly f_b . That signal is low-pass filtered and digitised by an analog-to-digital converter (ADC) at a modest rate — 8 MHz here, rather than the tens of gigahertz the raw RF would need. This is the central trick of FMCW: it converts a timing measurement that would need picosecond resolution into a frequency measurement that an ordinary ADC can make.

2.1.2 From beat frequency to a range profile

Measuring f_b is a spectral estimation problem, so the natural tool is the Fourier transform. Taking an N_s -point FFT across the N_s ADC samples of a single chirp produces a *range profile*: bin n of the FFT corresponds, through Equation (2.2), to range

$$R_n = n \cdot \frac{c}{2B}, \quad (2.3)$$

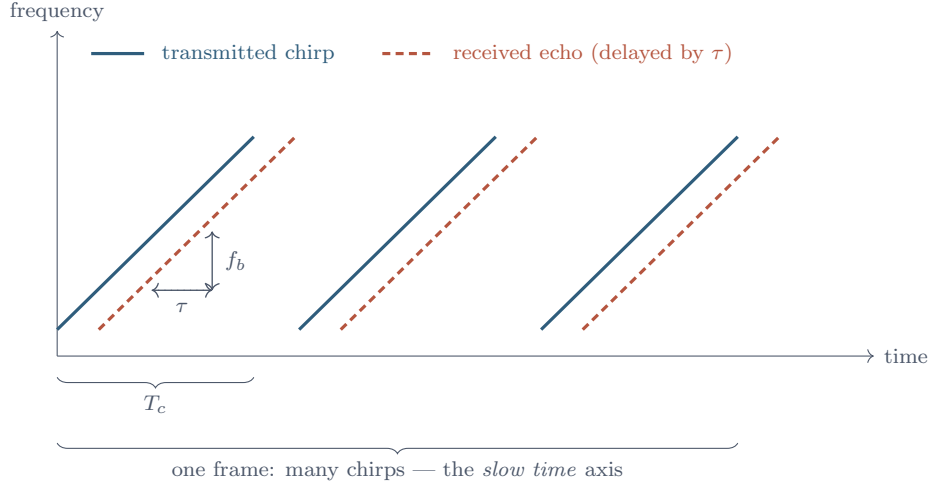


Figure 2.1: FMCW operation. Each chirp sweeps linearly in frequency. The echo is a delayed copy, so at any instant the transmitted and received frequencies differ by $f_b = S\tau$, proportional to target range. A frame consists of many such chirps in succession; the axis running across chirps is called *slow time* and is where Doppler information lives.

so the range resolution is

$$\Delta R = \frac{c}{2B} \approx 8.3 \text{ cm} \quad (2.4)$$

for the 1.798 GHz bandwidth used here. A peak in the range profile is an object. This first transform is called the **Range FFT**, and its output is the data this thesis compresses.

2.1.3 Slow time and Doppler

A single chirp gives range but not velocity. To get velocity, the radar transmits many chirps in rapid succession — a *frame* — and looks at how the *phase* of a given range bin evolves from chirp to chirp.

The reasoning is worth spelling out, because the whole of DRHE rests on it. A target at range R produces a range-FFT value whose phase is $\theta = -4\pi R/\lambda$, where λ is the wavelength. Over the chirp-to-chirp interval T , a target moving at radial velocity v changes range by vT , so its phase changes by

$$\Delta\theta = -\frac{4\pi vT}{\lambda}. \quad (2.5)$$

At 77 GHz, $\lambda \approx 3.9 \text{ mm}$, so even a very slow target produces a measurable phase step — which is exactly why radar is good at velocity. Critically for compression, $\Delta\theta$ is *constant* for a target at constant velocity. The phase of a given range bin therefore advances linearly across chirps, and its magnitude barely changes at all. That is a highly predictable sequence, and predictable sequences compress.

The axis running across chirps within a frame is called **slow time**, as opposed to the *fast time* axis of samples within a chirp. Taking a second FFT along slow time resolves $\Delta\theta$ into velocity: this is the **Doppler FFT**.

Carry forward: slow time is where the redundancy is. A target at constant velocity produces a range-bin sequence whose magnitude is nearly constant and whose phase advances by a constant increment. Both DRHE and LPC exist to exploit exactly this.

2.2 Angle, MIMO and the virtual array

2.2.1 Why more antennas

Range and velocity are not enough to drive a car. The radar also has to know *where* a target is in angle, and how well it can separate two targets at the same range and velocity but different bearings.

Angle is measured by comparing the phase of the same echo at physically separated receive antennas. A wavefront arriving at angle ϕ reaches adjacent antennas, spaced d apart, with a path difference $d \sin \phi$, hence a phase difference

$$\Delta\psi = \frac{2\pi d \sin \phi}{\lambda}. \quad (2.6)$$

An FFT across the receive-antenna axis — the **Angle FFT** — turns those phase differences into a bearing estimate. The angular resolution improves with the physical size of the array, which in practice means with the number of elements.

2.2.2 MIMO: buying antennas cheaply

Building a large physical receive array is expensive. Multiple-Input Multiple-Output (MIMO) radar gets the same effect more cheaply. If N_t transmitters and N_r receivers are arranged suitably, and the signal from each transmitter can be separated at each receiver, then each of the $N_t \times N_r$ transmit–receive pairs behaves like an element of a *virtual array*. The sensor used here has $N_t = 12$ and $N_r = 16$, giving 192 virtual elements from 28 physical antennas.

2.2.3 Time-division multiplexing, and why it matters here

The separability requirement has to be met somehow. The simplest method, and the one this sensor uses, is *time-division multiplexing* (TDM): the transmitters fire one at a time, in a fixed rotation, so whatever a receiver hears during transmitter t ’s slot must have come from transmitter t .

TDM is simple and robust, and it has one consequence that this thesis turns out to hinge on. Inside every repetition of the frame — every “chirp loop” — all N_t transmitters fire in sequence before the loop repeats. When the resulting data is laid out as a two-dimensional array of (ramp, range bin), which is how a streaming compressor sees it, the ramp index is

$$m = c N_t + t, \quad c \in [0, N_c), \quad t \in [0, N_t), \quad (2.7)$$

where c indexes the chirp loop and t the transmitter. The transmitter index varies *fastest*.

This is the correct physical time order. It is also, as Figure 2.2 makes visual, a trap. Ramp m and ramp $m - 1$ are different physical transmitters $N_t - 1$ times out of N_t . They are separated in *space* by the array baseline, not in *time* by a pulse interval, and the phase relationship between them is governed by Equation (2.6) — the target’s bearing — not by Equation (2.5) — the target’s velocity. The genuine slow-time neighbour of ramp m , the one whose phase differs by a Doppler step, is ramp $m - N_t$.

Carry forward: on a TDM-MIMO radar the ramp axis is transmitter-interleaved. Any algorithm that assumes consecutive ramps are consecutive *in time on the same antenna* — which includes DRHE as published, and a plain lag-1/lag-2 linear predictor — is operating on the wrong axis. This is the central finding of Chapter 6.

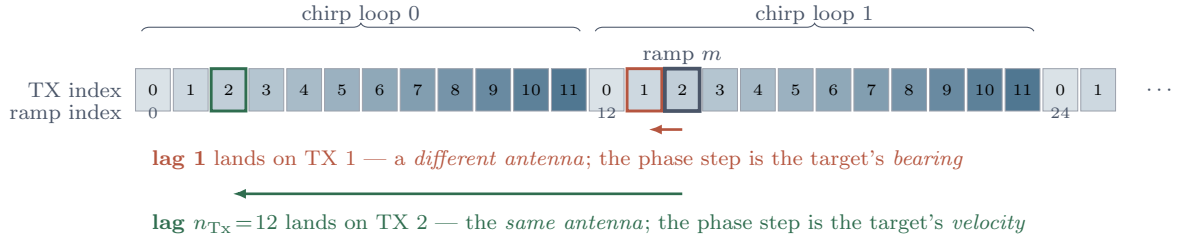


Figure 2.2: Why “the previous ramp” is the wrong neighbour on a TDM-MIMO cascade. Each cell is one ramp, labelled with the transmitter that produced it; the twelve transmitters rotate inside every chirp loop. Predicting ramp m from ramp $m - 1$ (red) crosses a transmitter boundary eleven times out of twelve, so the two ramps differ by the target’s array steering phase. Predicting from ramp $m - n_{\text{Tx}}$ (green) stays on the same physical antenna and therefore on the genuine Doppler axis.

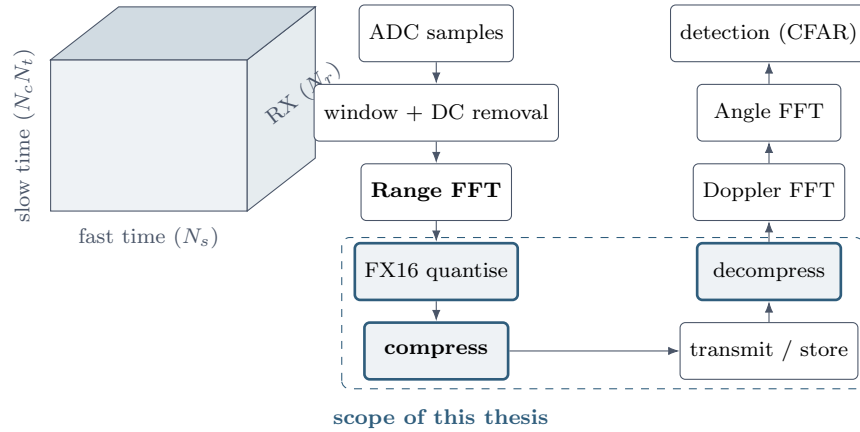


Figure 2.3: The radar data cube and the processing chain. The compressor sits immediately after range-FFT and fixed-point quantisation, before the data leaves the sensor. Everything downstream — Doppler FFT, angle FFT, detection — sees the reconstructed data, which for a lossless compressor is bit-identical to the original.

2.3 The radar data cube, and why it is large

Putting the three axes together gives the *radar data cube*: fast time (samples within a chirp) $\times$ slow time (ramps) $\times$ receive channel. Three FFTs are applied along the three axes in turn, yielding range, velocity and angle. Figure 2.3 shows the arrangement and where in the chain this thesis intervenes.

The volume follows directly. For the sensor used here, one frame is

$$N_s \times N_c \times N_r \times N_t = 256 \times 16 \times 16 \times 12 = 786,432 \quad (2.8)$$

complex samples. Each is two 16-bit integers, so a frame is 3,145,728 B — exactly 3 MiB. At a typical frame rate this is gigabits per second from a single sensor, and a vehicle carries several.

2.4 Lossless compression: how it actually works

2.4.1 The two categories

Compression divides into *lossless*, where the decompressed data is bit-identical to the input, and *lossy*, where some information is permanently discarded in exchange for a smaller payload [14]. Lossy compression can reach far higher ratios; lossless compression is bounded by the information content of the data itself. Section 1.3 explained why this application needs the lossless kind.

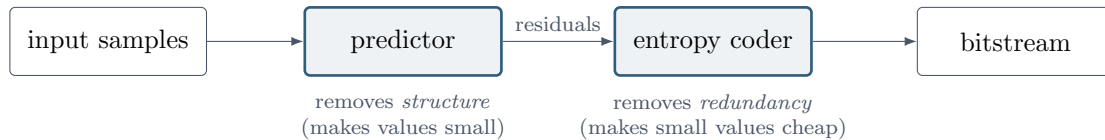


Figure 2.4: The standard structure of a lossless compressor. The two stages do different jobs and can fail independently — which is precisely what Chapter 6 finds had happened.

2.4.2 The bound: entropy

There is a hard limit on lossless compression, and knowing it is what makes the results in Chapter 6 interpretable. If a source emits symbols x with probabilities $p(x)$, its *order-0 entropy* is

$$H = - \sum_x p(x) \log_2 p(x) \quad \text{bits per symbol}, \quad (2.9)$$

and no code that treats symbols independently can average fewer bits than H [14]. This gives a concrete yardstick: measure H for the data, compare it against what the coder actually achieves, and the gap is the coder’s inefficiency. Section 6.4 does exactly this, and finds the entropy coder used here runs only 6.2% above the order-0 entropy of its input — which is good, and which means the room for improvement is not in the entropy coder.

2.4.3 The standard two-stage structure

Almost every practical lossless compressor has the same shape (Figure 2.4): a *decorrelation* stage followed by an *entropy coding* stage.

The **decorrelation** stage exploits structure. If consecutive samples are similar, predict each one from its neighbours and transmit only the *residual*, the difference between prediction and truth. When the prediction is good, residuals cluster tightly around zero even though the original samples spanned the full dynamic range. Nothing is lost: given the residual and the same prediction rule, the decoder reconstructs the original exactly.

The **entropy coding** stage exploits the resulting skew in the distribution. Values that occur often get short codewords; values that occur rarely get long ones. Huffman coding [15] is the classic construction and produces the optimal prefix-free code for a known symbol distribution.

Crucially, the two stages are independent. A predictor that fails — that produces residuals no more skewed than its input — does not break the coder; the coder simply carries on compressing whatever it is handed, and the compression ratio can still look respectable. Detecting this requires deliberately measuring what the coder achieves *without* the predictor, which is not a standard practice and which is how the central finding of this thesis came to light.

2.4.4 Open-loop versus closed-loop prediction

One subtlety in predictive coding matters enough to state now, because it caused a defect described in Section 5.4.1.

The encoder predicts from past samples. So does the decoder — but the decoder does not have the past *original* samples, only the past *reconstructed* ones. In *open-loop* prediction the encoder uses the originals; in *closed-loop* prediction (classically, differential pulse-code modulation, DPCM) the encoder reconstructs each sample exactly as the decoder will, and predicts from that.

When compression is perfectly lossless the two are identical, because reconstruction equals the original. The moment anything — a saturation, a rounding, a clamp — makes

Table 2.1: ColoRadar cascade capture parameters used throughout this thesis.

Parameter	Symbol	Value
ADC samples per chirp	N_s	256
Chirp loops per frame	N_c	16
Receive channels	N_r	16
Transmit channels (TDM)	N_t	12
Ramps per frame	$N_c N_t$	192
Fast-time sample rate	f_s	8 MHz
Chirp bandwidth	B	1.798 GHz
Chirp-to-chirp interval	T_c	78.5 μ s
Centre frequency	f_c	77 GHz
Range resolution	ΔR	8.34 cm
Frame size (raw)	—	3,145,728 B

reconstruction differ from the original by even one bit, an open-loop encoder and its decoder begin predicting from different histories and *diverge*. With an IIR predictor, as in DRHE, that divergence is then carried forward through every subsequent ramp. Closing the loop confines the error to the affected sample.

Carry forward: a compression ratio is the product of two independent effects, and a healthy-looking ratio does not imply a working predictor. Always measure the entropy coder alone as a baseline.

2.5 The sensor and the dataset

2.5.1 AWR2243 and the MMWCAS-RF-EVM

The radar front end assumed throughout is the Texas Instruments AWR2243, a 76–81 GHz FMCW transceiver with three transmit and four receive channels [13]. Importantly for this work, the AWR2243 performs no signal processing of its own: it transmits, receives, mixes and digitises, then streams raw ADC samples out over a MIPI CSI-2 interface. It therefore exposes exactly the payload that creates the transport bottleneck.

Four AWR2243 devices are cascaded on the MMWCAS-RF-EVM evaluation module, giving 12 transmit and 16 receive channels on one synchronised board — the 192-element virtual array of Section 2.2.

2.5.2 ColoRadar

Rather than synthetic stimulus, this thesis uses real captures from the public ColoRadar dataset [12], which includes cascade recordings from exactly this module across indoor, outdoor and subterranean environments. Table 2.1 lists the relevant parameters. Fifty consecutive frames were used throughout; that is 157,286,400 complex samples, or 150 MiB of raw data, which is enough to make a claim of losslessness meaningful.

Each complex sample is stored as two contiguous `int16` values. The index of the sample at (fast-time s , chirp loop c , receiver r , transmitter t) in the file is

$$i = 2(s + N_s(c + N_c(r + tN_r))), \quad (2.10)$$

so the layout from slowest to fastest varying is $[t, r, c, s]$. Reshaping this into the (fast time, ramp, receiver) cube the processing chain expects is what produces the transmitter-interleaved ramp axis of Equation (2.7).

2.6 Number representation and quantisation

2.6.1 Fixed point versus floating point

Hardware has to represent numbers somehow, and the choice is consequential.

Fixed-point representation allocates a constant number of bits to the integer and fractional parts, commonly written *Qm.n*. Because the binary point never moves, addition, subtraction and multiplication are ordinary integer operations: cheap, single-cycle, and small in silicon [16]. The price is a fixed dynamic range. Values too large wrap around; values too small round to zero.

Floating-point representation stores a sign, exponent and mantissa, so the binary point moves with the magnitude [17]. This buys enormous dynamic range but costs a great deal: every operation requires exponent alignment, normalisation and special-case handling, which needs a dedicated floating-point unit and takes multiple clock cycles. Chapter 6 shows this concretely — the DRHE implementation’s LUT count is dominated by four floating-point transcendental functions per sample, and they are why it misses its clock target.

2.6.2 Quantisation error, and the FX16 step

The range-FFT output is naturally a complex floating-point quantity. Before it can be compressed by a fixed-point pipeline it has to be quantised to 16-bit integers — the **FX16** step. This is a scaling followed by rounding:

$$\sigma = \frac{32767}{A_{\max} 2^{\text{shift}} \sum_k w_k / (2N_s)}, \quad x_{\text{FX16}} = \text{round}(\sigma \cdot x), \quad (2.11)$$

where $A_{\max}$ is the full-scale ADC amplitude and w is the window applied along fast time.

Two things about Equation (2.11) matter later. First, the rounding introduces an error bounded by half a least-significant bit, which acts as a wideband noise floor [18]; this is the dominant source of detection degradation in the whole chain, and Chapter 6 separates it cleanly from the cost of compression. Second, and more consequentially for this thesis, the scale factor is *fixed* — it is calibrated from a theoretical full-scale ADC sinusoid, not from the data. If the actual capture sits well below full scale, the top bits of every resulting **int16** are structurally zero. Section 6.4 shows that this is exactly what happens, and that it accounts for more of the reported compression ratio than the compression algorithms do.

Carry forward: the FX16 scale factor is fixed and calibrated to full scale. Whether the capture actually fills the container is a property of the sensor’s gain staging, and it silently inflates any compression ratio measured downstream.

2.7 FPGAs and high-level synthesis

2.7.1 What an FPGA is

A field-programmable gate array is a chip full of small configurable logic blocks and configurable wiring between them. Rather than executing instructions, an FPGA *becomes* the circuit you describe. The practical consequences for this application are three:

- **Parallelism.** A processor handles four receive channels one after another. An FPGA can instantiate four copies of the datapath and do all four in the same clock cycle. Every design in this thesis does exactly that.
- **Deterministic latency.** There is no cache, no operating system, no interrupt jitter. A design that takes 61,253 cycles takes 61,253 cycles every time, which is what a safety-critical pipeline needs.

- **Reconfigurability.** The design can be changed after fabrication, which is what makes an FPGA the right platform for a thesis — and what made it practical to build, measure and then substantially revise two compressors inside one project.

In production, a design proven on an FPGA is normally migrated into an ASIC or into a fixed-function accelerator block inside a larger SoC, for cost and power reasons [19, 21]. The FPGA here is a proof of concept, not a proposed product.

2.7.2 The four currencies: LUT, FF, BRAM, DSP

FPGA resource usage is reported in four units, and it is worth knowing what each is because the two algorithms in this thesis trade them against each other in an interesting way.

LUT

Look-up table. A small memory that implements an arbitrary Boolean function of a few inputs. LUTs are the raw material of combinational logic. Complicated arithmetic — square roots, arctangents — costs a lot of LUTs.

FF *Flip-flop.* A one-bit register. Flip-flops hold values between clock cycles; deep pipelines need many of them.

BRAM

Block RAM. Dedicated on-chip memory, here counted in 18 kbit units. Arrays that are too large to live in flip-flops go here. BRAMs typically have two access ports, which becomes important below.

DSP

DSP slice. A hardened multiply–accumulate unit. Using a DSP for a multiply is far cheaper than building one from LUTs.

The target device throughout is an AMD Kintex UltraScale+ `xcku5p-ffvb676-2-e`, which provides 216,960 LUTs, 433,920 flip-flops, 960 18 kbit BRAMs and 1,824 DSP slices.

2.7.3 Pipelining, initiation interval and $F_{\max}$

Three timing concepts recur throughout the results.

Pipelining splits a long computation into stages separated by registers, so several inputs can be in flight at once. The *depth* of a pipeline is how many cycles a single input takes to traverse it.

Initiation interval (II) is how many cycles must elapse between accepting one input and the next. $II = 1$ means a new sample every cycle, which is the goal. $II = 2$ halves throughput. II is limited by the tightest resource constraint in the loop body — very often, as Chapter 6 shows, by the number of access ports on a block RAM.

$F_{\max}$ is the highest clock frequency at which the longest combinational path still settles within one period. A design that estimates 81.88 MHz cannot be clocked at 100 MHz without either redesign or a slower clock.

Latency, throughput and $F_{\max}$ interact: a frame of K beats at initiation interval II and clock f takes roughly $K \cdot II / f$ seconds, *plus* the pipeline depth once per invocation. That last term is usually negligible — and Section 5.5 contains a case where it emphatically is not, costing 65,000 cycles that a naive model misses entirely.

2.7.4 High-level synthesis

Traditionally FPGA designs are written in a hardware description language such as VHDL or Verilog, at the level of registers and wires. *High-level synthesis* (HLS) instead takes annotated C++ and generates the RTL automatically. Loops become state machines; arrays become

memories; pragmas tell the tool to pipeline a loop, partition an array, or expose a port as an AXI4-Stream interface.

HLS is used throughout this thesis for a specific reason beyond convenience: because the design source is C++, the *same source* can be compiled and run as ordinary software against the MATLAB reference, before any hardware is generated. That makes the verification ladder of Chapter 4 possible, and it is how both defects in Section 5.4 were caught in seconds rather than in hours of RTL simulation.

2.7.5 AXI4-Stream

The interface used by both compressors is AXI4-Stream: a simple valid/ready handshake carrying a fixed-width data word per beat. The designs here accept 128 bits per beat — four receive channels, each a 16-bit real and a 16-bit imaginary part — and emit 256 bits per beat of compressed bitstream. A useful consequence for benchmarking is that “128 bits per cycle” is the theoretical maximum input rate, and therefore what a perfectly efficient design at $\Pi = 1$ would achieve. Section 7.1 uses this to resolve an ambiguity in a published throughput figure.

2.8 Related work and the baseline

2.8.1 Deterministic industrial schemes

Because automotive safety systems demand bounded latency, commercial radar processors use lightweight deterministic compressors. Texas Instruments uses Exponential-Golomb Encoding [20]; NXP’s S32R family uses block floating-point formats such as CP4D, in which several mantissas share one exponent [21]. These are described as “near lossless” — CP4D introduces under 0.25 dB of signal loss — and reach ratios of roughly 2 to 4.

2.8.2 Block Adaptive Quantization

BAQ originates in synthetic aperture radar [22]. It divides the data into blocks, estimates each block’s statistics, and adapts the quantiser step size accordingly, so weak blocks are quantised finely and strong blocks coarsely. It is lossy. It appears in this thesis as a comparison point, and Chapter 6 shows what its losses cost in detection terms.

2.8.3 Neural and spectral approaches

Autoencoder-based compressors reach high ratios but are expensive to deploy at the sensor [23]. AdaRadar [24] avoids this by keeping the edge side simple — a discrete cosine transform and spectral pruning — and placing the neural decoder at the SoC, with a feedback loop that adapts the rate to downstream perception performance. These are outside the scope of this thesis, which concerns itself with lossless schemes, but they define the upper end of the ratio spectrum.

2.8.4 The baseline: Kiem’s thesis

The direct predecessor of this work is Lukas Kiem’s master’s thesis [11], which performed a design-space exploration across Exponential-Golomb variants, Infineon common-exponent formats, and predictive coders, and concluded that Doppler Redundancy Hybrid Encoding gave the best combination of compression ratio, noise-floor impact and detection accuracy. He implemented DRHE on a Zynq UltraScale+ device, reporting a compression ratio of 3.2, an IP-core latency of 230 ns and a throughput of “11.92 Gbit/s” at 100 MHz.

That thesis is the source of the DRHE specification used here, including the filter coefficients and the Huffman dictionary, both of which were verified entry-by-entry against it

(Section 3.1.6). It left three gaps — synthetic-only validation, a narrow algorithm comparison, and a compression-only implementation — which Section 1.4 listed as this thesis’s objectives.

It also, as Chapter 6 shows, contains an assumption that does not survive the move to a TDM-MIMO cascade. That is not a criticism of the work: a single-transmitter radar has no transmitter-interleaved ramp axis, and on such a radar the specification is correct as written. It becomes wrong only when applied, unmodified, to a sensor of a different kind — which is exactly what this thesis did before measuring it.

2.9 Summary of what the later chapters assume

1. The data to be compressed is the *range-FFT output*, quantised to 16-bit integers, arranged as (ramp $\times$ range bin $\times$ receiver).
2. The redundancy worth exploiting lives along *slow time*: a target at constant velocity gives nearly constant magnitude and a constant phase increment.
3. On this sensor the ramp axis is *transmitter-interleaved*, so the slow-time neighbour of ramp m is ramp $m - N_t$, not ramp $m - 1$.
4. A compression ratio is the product of a decorrelation gain and an entropy-coding gain, and the two must be measured separately.
5. The FX16 scale factor is fixed to full scale, so container occupancy is a confounding variable in any measured ratio.
6. FPGA cost is measured in LUTs, flip-flops, BRAMs and DSPs, and timing in initiation interval, pipeline depth and $F_{\max}$.

Chapter 3

The Two Algorithms

Both compressors follow the two-stage structure of Section 2.4: a predictor that turns samples into small residuals, followed by an entropy coder that turns small residuals into few bits. They share the entropy coder exactly, which is deliberate — it means any difference in compression ratio between them is attributable entirely to the predictor, and it is what makes the no-prediction baseline of Chapter 6 a fair comparison for both.

This chapter specifies the entropy coder first, then each predictor.

3.1 The shared entropy coder: turning residuals into bits

This section is deliberately detailed. The entropy coder is where the compression ratio is physically realised — it is the only part of either design that actually emits bits — so understanding exactly what it charges for is what makes the results of Chapter 6 interpretable rather than merely quotable.

3.1.1 What a residual is

Let $x[m]$ be the sample to be transmitted and let P be some rule — the *predictor* — that produces an estimate

$$\hat{x}[m] = P(x[m-1], x[m-2], \dots) \quad (3.1)$$

from samples that have *already been transmitted*. The encoder does not send $x[m]$. It sends the *residual*

$$d[m] = x[m] - \hat{x}[m]. \quad (3.2)$$

The decoder, having already reconstructed $x[m-1], x[m-2], \dots$, runs the identical rule P on them, obtains the identical $\hat{x}[m]$, and recovers

$$x[m] = d[m] + \hat{x}[m]. \quad (3.3)$$

Two things follow, and both matter.

Nothing is lost. Equation (3.2) is a *re-expression*, not an approximation. Given $\hat{x}[m]$, the map $x \mapsto d$ is a bijection. The sequence $\{d[m]\}$ carries exactly the information the sequence $\{x[m]\}$ carried — no more, no less. This is why a predictive coder can be lossless at all.

The predictor must be causal. P may only use samples the decoder already has. This is not a stylistic preference but a hard constraint: a predictor that peeked at $x[m]$ itself would be unreproducible at the far end. It is also why Section 2.4.4 makes such a point of *which* version of the past the encoder uses — the originals or the reconstructions — since those differ the moment anything is not exactly lossless.

Table 3.1: The same sixteen real samples, written two ways. Range bin 75, receive channel 1, ramps 168–183. Both rows carry identical information.

m	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183
x (raw)	−69	−52	59	−6	−29	−15	59	−20	−69	−73	−45	84	−73	−50	59	−5
d (residual)	3	−3	0	−1	1	3	−4	3	−1	−2	−1	0	−2	4	−1	1

3.1.2 Why a re-expression can be worth anything

If no information is lost and none is gained, why is the residual sequence any cheaper to write down?

Because the *distribution* has changed, and a well-chosen code charges by distribution. Table 3.1 shows this on real data: one range bin of one receive channel, sixteen consecutive ramps, taken from frame 0 of the ColoRadar capture.

The raw values swing over $[-73, +84]$; writing any of them needs eight bits. The residuals never leave $[-4, +4]$; four bits would do. Since the two rows are informationally equivalent, that factor of two is free — provided the coder can be told about it.

And there is the catch. The bound $[-4, +4]$ holds for *these* sixteen samples. Somewhere else in the frame a target appears, or the scene changes, and a residual of 400 arrives. A fixed four-bit field cannot carry it. What is needed is a code that spends few bits on the common case, many on the rare one, and lets the decoder tell which is which.

3.1.3 The parsing problem

The naive approach fails immediately. Suppose the rule were “small values get four bits, large values get twelve”. When the decoder reads four bits, it has no way to know whether it has just read a complete small value or the first third of a large one. Variable-length coding is not merely a matter of writing fewer bits; it is a matter of writing them so that the boundaries can be recovered.

There are two standard answers:

1. **A prefix-free code over every possible value.** Assign every one of the 65,536 possible 16-bit residuals its own codeword, no codeword being a prefix of another. This is optimal, and it is what a Huffman code over the raw alphabet would give — but it needs a 65,536-entry table, and the table has to be derived from the data and transmitted alongside it.
2. **Split the value into a size and a payload.** Send a small *description of how big the value is*, coded with a prefix-free code over a handful of symbols, followed by the value itself in exactly that many raw bits. The decoder reads the size first, so it always knows how many bits to read next.

Both compressors here take the second route, which is the same construction JPEG uses for DC coefficients. It is cheap in every respect that matters for hardware: the prefix-free table has sixteen entries rather than 65,536, it can be fixed at design time rather than derived per frame, and decoding a symbol is a sixteen-way comparison rather than a tree walk.

3.1.4 The category S_4

The “size” half of the split is the category

$$S_4(v) = \begin{cases} 0 & v = 0, \\ \min(15, \lfloor \log_2 |v| \rfloor + 1) & v \neq 0, \end{cases} \quad (3.4)$$

Table 3.2: The first few categories. Category $k \geq 1$ covers $|v| \in [2^{k-1}, 2^k - 1]$, which counting both signs is exactly 2^k distinct values — precisely what k raw bits can enumerate.

S_4	values covered	count	
0	$v = 0$	1	the only category with no payload
1	$v \in \{-1, +1\}$	2	
2	$ v \in \{2, 3\}$	4	
3	$ v \in [4, 7]$	8	
4	$ v \in [8, 15]$	16	
$\vdots$			
15	$ v \in [16384, 32767]$	32,768	exactly fills 15 bits

which is simply the number of bits needed to write $|v|$. Its ranges are given in Table 3.2.

The design is tight by construction: category k contains exactly 2^k values, so k payload bits identify one of them with nothing wasted and nothing left over. Section 3.1.12 shows what happens at the one place where “nothing left over” turns out to be a problem.

3.1.5 The APPEND bits

The payload half is

$$A(v) = \begin{cases} v & v \geq 0, \\ v + 2^{S_4(v)} - 1 & v < 0, \end{cases} \quad (3.5)$$

written in $S_4(v)$ raw bits.

The offset for negative values looks arbitrary but is doing something specific. Table 3.3 enumerates category 3 in full.

Table 3.3: Every codepoint of category 3. The negative values map to patterns with a clear most significant bit, the positive values to patterns with it set — so the sign is recovered from that one bit without transmitting a sign bit.

v	$A(v)$	3-bit pattern	MSB
-7	0	000	0
-6	1	001	0
-5	2	010	0
-4	3	011	0
+4	4	100	1
+5	5	101	1
+6	6	110	1
+7	7	111	1

Because positive values in category k span $[2^{k-1}, 2^k - 1]$ — exactly the k -bit patterns with the top bit set — and the offset maps the negatives onto $[0, 2^{k-1} - 1]$ — exactly those with it clear — the sign falls out of the payload for free. The decoder’s rule is therefore

$$v = \begin{cases} A & \text{if the MSB of the } S_4 \text{ payload bits is set,} \\ A - (2^{S_4} - 1) & \text{otherwise,} \end{cases} \quad (3.6)$$

which in hardware is one bit test and one subtract.

3.1.6 Cost, and the shape of the dictionary

The total cost of coding one residual is

$$L(v) = \underbrace{\ell(S_4(v))}_{\text{Huffman codeword}} + \underbrace{S_4(v)}_{\text{payload}}, \quad (3.7)$$

where $\ell(\cdot)$ is the codeword length for that category. The dictionary is *fixed* — not computed per frame — which saves transmitting it and removes a serial dependency from the hardware. Its lengths are

$$\ell = [4, 3, 2, 2, 2, 5, 6, 7, 8, 9, 10, 11, 14, 14, 13, 12] \text{ bits.} \quad (3.8)$$

The full dictionary, codewords included, is in Appendix A. Every one of the sixteen entries was verified against the published table [11, Table A.1]; all sixteen agree once the published most-significant-bit-first codes are reversed for the least-significant-bit-first packer used in this implementation.

Substituting into Equation (3.7) gives the cost curve that governs everything downstream:

S_4	$ v $ range	ℓ	payload	total
0	0	4	0	4
1	1	3	1	4
2	2–3	2	2	4
3	4–7	2	3	5
4	8–15	2	4	6
5	16–31	5	5	10
6	32–63	6	6	12
7	64–127	7	7	14
8	128–255	8	8	16

Three features of this curve deserve attention.

Everything below 16 is nearly free. A residual anywhere in $[-15, +15]$ costs between four and six bits. Against sixteen raw bits that is already a compression ratio between 2.7 and 4.0, achieved without the predictor doing anything clever — it only has to keep residuals small.

There is a cliff at category 5. Cost jumps from six bits to ten. A residual of 15 costs six bits; a residual of 16 costs ten. The coder is, by design, forgiving of small errors and severe about larger ones.

Above category 8 the code loses. A residual of 128 costs sixteen bits — exactly what the raw value would have cost. Beyond that the coder actively expands. This is not a flaw: it is the inevitable consequence of assigning short codewords to some symbols, and it is why the `random-int16` edge case of Section 4.4 expands the data by 67%.

The dictionary’s shape encodes an assumption about the data — that categories 2, 3 and 4 dominate, since those get the two-bit codewords. Section 3.1.10 measures whether that assumption holds.

3.1.7 Worked example 1: one value, in full

Take $v = -5$.

1. $|v| = 5$, and $\lfloor \log_2 5 \rfloor + 1 = 3$, so $S_4 = 3$.
2. $v < 0$, so $A(v) = -5 + 2^3 - 1 = 2$, written in three bits as 010.

3. The dictionary gives category 3 the codeword 00, two bits.
4. The emitted bits are 00 010 — five bits, against sixteen for the raw value.

Decoding runs the same steps backwards. The decoder matches 00 against the dictionary, recovers $S_4 = 3$, reads the next three bits 010, sees the most significant is 0 so the value was negative, and computes $2 - (2^3 - 1) = -5$.

3.1.8 Worked example 2: sixteen real samples, both ways

The single-value example shows the mechanism. The following example shows why it matters, and it doubles as a concrete demonstration of the central result of this thesis.

Table 3.4 traces sixteen consecutive ramps of one range bin — the same bin as Table 3.1 — through the DRHE predictor twice. The left half predicts from ramp $m - 1$, which is how the algorithm is described in the literature. The right half predicts from ramp $m - n_{\text{Tx}}$, which Section 2.2.3 argued is the true slow-time neighbour. Everything else — the filters, the coefficients, the entropy coder, the dictionary — is identical between the two halves. Only the lag differs.

Table 3.4: Sixteen real samples traced through DRHE at two prediction lags. Range bin 75, receive channel 1, frame 0, ramps 168–183; real part shown, the imaginary part behaves identically. The window is late enough in the frame that both predictors’ filters are fully converged. The two halves differ only in which earlier ramp the prediction comes from.

m	TX	x	predicted from ramp $m - 1$				predicted from ramp $m - n_{\text{Tx}}$			
			$\hat{x}$	d	S_4	bits	$\hat{x}$	d	S_4	bits
168	0	-69	67	-136	8	16	-72	3	2	4
169	1	-52	-75	23	5	10	-49	-3	2	4
170	2	59	81	-22	5	10	59	0	0	4
171	3	-6	64	-70	7	14	-5	-1	1	4
172	4	-29	24	-53	6	12	-30	1	1	4
173	5	-15	-29	14	4	6	-18	3	2	4
174	6	59	-6	65	7	14	63	-4	3	5
175	7	-20	-83	63	6	12	-23	3	2	4
176	8	-69	-93	24	5	10	-68	-1	1	4
177	9	-73	-65	-8	4	6	-71	-2	2	4
178	10	-45	-44	-1	1	4	-44	-1	1	4
179	11	84	28	56	6	12	84	0	0	4
180	0	-73	70	-143	8	16	-71	-2	2	4
181	1	-50	-75	25	5	10	-54	4	3	5
182	2	59	81	-22	5	10	60	-1	1	4
183	3	-5	67	-72	7	14	-6	1	1	4
total bits						176	66			
bits per value						11.00	4.12			
compression ratio						1.455	3.879			

The left half is a predictor that is barely working. Its residuals (-136 , $+23$, -22 , -70 , $\dots$) are frequently *larger in magnitude than the samples they replace*, they land in categories 5 through 8, and they cost 11.00 bits per value — a compression ratio of 1.455, which is to say the entropy coder is salvaging a little from a predictor that has actively made things worse.

The right half is the same predictor on the correct axis. Its residuals never exceed 4 in magnitude, they land in categories 0 through 3, and they cost 4.12 bits per value.

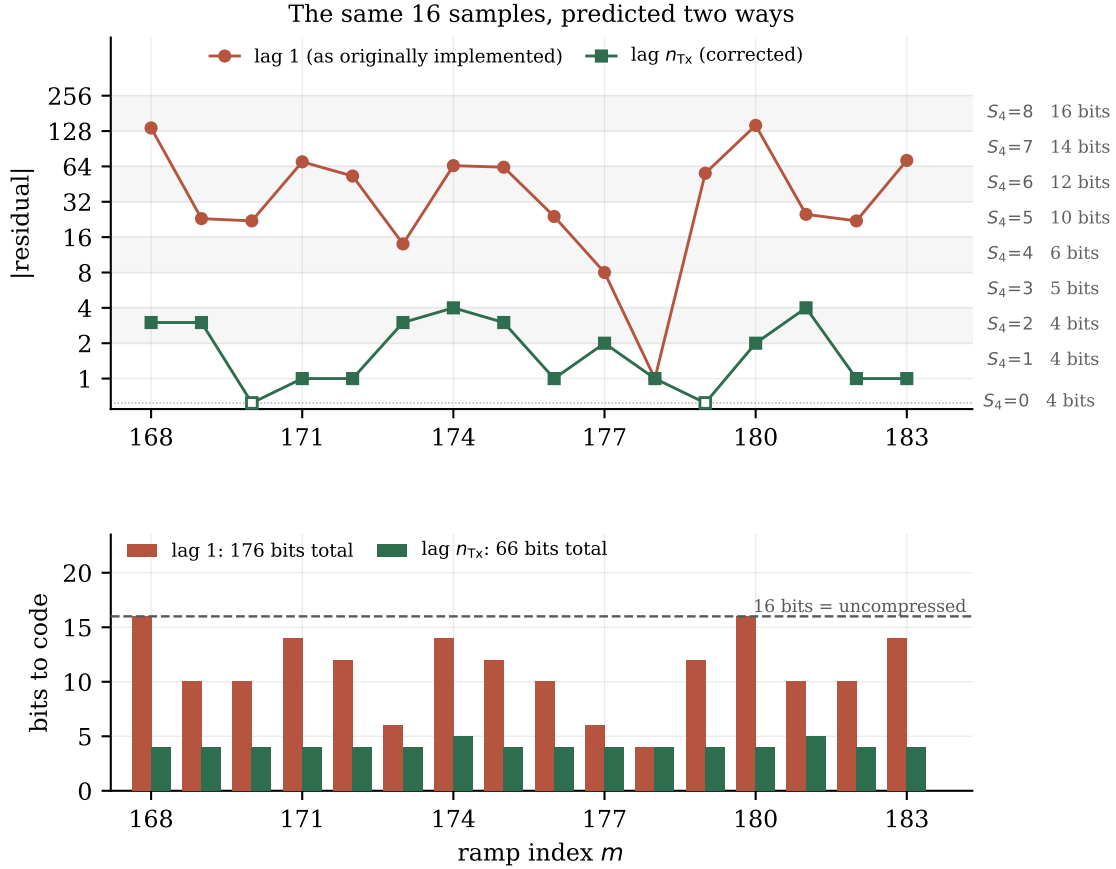


Figure 3.1: Table 3.4 drawn. Above: residual magnitude on a logarithmic scale, with the S_4 category bands and their bit costs marked on the right; hollow markers are residuals of exactly zero. Below: the bits each sample costs, against the 16 bits it would have taken uncompressed. The lag-1 predictor spends 176 bits on this window; the same predictor on the correct axis spends 66.

Why the difference is so large

Table 3.5 answers this directly, and it is worth studying because it is the mechanism behind every result in Chapter 6. Recall from Section 2.1 that DRHE predicts magnitude with a smoother and phase with a double integrator, both of which work when the quantity being predicted changes slowly and smoothly.

Look at the $\Delta\theta$ columns. Against ramp $m - 1$ the phase increment is -2.244 , then $+0.592$, then $+1.230$, then -0.678 — it swings over most of $[-\pi, \pi]$ with no pattern a two-tap filter could follow. Against ramp $m - n_{Tx}$ it is 0.002 , -0.040 , 0.017 , -0.008 : essentially zero, frame after frame, because this is a stationary target and twelve ramps later the same antenna sees the same phase.

The magnitude column tells the same story: ± 22 against the immediate predecessor, ± 4 against the same-transmitter predecessor.

The reason is the one given in Section 2.2.3. Ramps 168 and 167 were transmitted by *different antennas*, so the phase between them is set by where the target is in *angle*, via Equation (2.6) — and the twelve transmitters are at twelve different positions, so that phase jumps around on a period-12 pattern. Ramps 168 and 156 were transmitted by the *same* antenna, so the phase between them is set by how fast the target is *moving*, via Equation (2.5) — and it is not moving.

Table 3.5: The same window in polar coordinates. Against ramp $m - 1$ the magnitude moves by up to 22 and the phase jumps unpredictably over the full $\pm\pi$ range. Against ramp $m - n_{\text{Tx}}$ the magnitude moves by at most 4 and the phase increment is essentially zero — a stationary target, exactly what the double integrator is built for.

m	TX	sample		change vs. $m - 1$		change vs. $m - n_{\text{Tx}}$	
		$ x $	$\angle x$	$\Delta x $	$\Delta\theta$	$\Delta x $	$\Delta\theta$
168	0	74.1	-2.769	-21.8	-2.244	-3.2	0.002
169	1	91.3	-2.177	17.2	0.592	-0.0	-0.040
170	2	101.0	-0.947	9.8	1.230	-2.5	0.017
171	3	110.2	-1.625	9.1	-0.678	3.0	-0.008
172	4	107.0	-1.845	-3.2	-0.220	-2.2	0.004
173	5	112.0	-1.705	5.0	0.140	-0.3	0.018
174	6	97.0	0.917	-15.0	2.622	-0.4	0.023
175	7	81.5	1.819	-15.5	0.902	0.4	-0.027
176	8	73.4	2.794	-8.1	0.975	-0.1	0.043
177	9	75.0	-2.913	1.6	0.576	2.4	0.021
178	10	95.3	-2.063	20.3	0.850	0.9	0.005
179	11	97.8	-0.537	2.5	1.526	1.9	-0.013
180	0	77.5	-2.799	-20.3	-2.263	3.4	-0.031
181	1	92.6	-2.141	15.2	0.659	1.4	0.036
182	2	102.6	-0.958	10.0	1.182	1.6	-0.011
183	3	106.1	-1.618	3.5	-0.659	-4.0	0.007

The actual bits

It is worth seeing the bitstream itself. Coding the first six residuals of the corrected column, using Equations (3.4) and (3.5) and the dictionary of Appendix A:

d	S_4	$A(d)$	codeword	payload	emitted	bits
+3	2	3	10	11	10 11	4
-3	2	0	10	00	10 00	4
0	0	—	0101	—	0101	4
-1	1	0	001	0	001 0	4
+1	1	1	001	1	001 1	4
+3	2	3	10	11	10 11	4

which concatenates to the 24-bit stream

1011 1000 0101 0010 0011 1011

carrying six samples that would have occupied 96 bits raw.

The same six samples on the uncorrected axis — residuals -136, +23, -22, -70, -53, +14 — fall into categories 8, 5, 5, 7, 6 and 4 and occupy $16 + 10 + 10 + 14 + 12 + 6 = 68$ bits. Nearly three times as many, for identical information.

3.1.9 Decoding the same stream

Decoding is the mirror image, and seeing it done confirms that no side information beyond the fixed dictionary is needed. The decoder holds a bit buffer and consumes from the low end.

1. Buffer is 1011 1000. . . . Match the low bits against the dictionary. The two-bit prefix 10 is category 2's codeword; no shorter or longer entry can also match, because the reversed code set is prefix-free. Consume two bits.

Table 3.6: Where the bits go, one frame, four receive channels, 196,608 values. “% values” is how often a category occurs; “% bits” is how much of the output it accounts for. The two columns are very different, and the difference is the whole point of entropy coding.

S_4	cost	no prediction		DRHE, lag 1		DRHE, lag n_{Tx}	
		% values	% bits	% values	% bits	% values	% bits
0	4	16.71	14.61	11.78	9.81	18.85	17.87
1	4	28.57	24.97	22.01	18.33	32.94	31.22
2	4	31.45	27.49	31.95	26.60	34.40	32.60
3	5	14.66	16.02	22.21	23.12	11.31	13.40
4	6	4.31	5.65	6.46	8.07	1.49	2.12
5	10	1.75	3.82	2.36	4.92	0.50	1.18
6	12	1.15	3.01	1.43	3.58	0.29	0.82
7	14	1.14	3.47	1.09	3.17	0.18	0.59
8	16	0.19	0.66	0.65	2.17	0.03	0.13
9	18	0.07	0.29	0.06	0.23	0.01	0.05
bits/value		4.576		4.803		4.220	
compression ratio		3.497		3.331		3.792	

- Category 2 means two payload bits. Read 11, so $A = 3$. The most significant of those two bits is set, so the value is positive: $d = +3$. Consume two bits.
- Add the prediction: the decoder has already reconstructed ramp $168 - 12 = 156$ for this bin, runs the identical filters on it, obtains $\hat{x} = -72$, and outputs $x = 3 + (-72) = -69$ — which is the value in Table 3.4.
- Buffer is now 1000 0101 Match 10 again: category 2. Read payload 00, so $A = 0$, MSB clear, so $d = 0 - (2^2 - 1) = -3$. Prediction -49 , output -52 . Correct.
- Buffer is 0101 0010 Now 10 does *not* match, because the low two bits are 01, which is no entry’s codeword. The four-bit prefix 0101 is category 0’s codeword. Category 0 has no payload and means $d = 0$ exactly. Prediction 59, output 59. Correct.

Step 5 is the one that shows why prefix-freeness is doing real work: the decoder never had to be told how long the next codeword would be.

3.1.10 Where the bits actually go

The sixteen-sample window is illustrative but small. Table 3.6 gives the same accounting over an entire frame — 196,608 values, all four receive channels — which is what the compression ratio actually integrates.

Several things are visible here that the headline ratios hide.

The expensive categories are rare but not negligible. Under lag-1 prediction, categories 5 through 8 account for only 5.53% of values but 13.84% of bits. One value in eighteen is consuming one bit in seven. That asymmetry is exactly why the tail of the residual distribution matters so much, and why Figure 6.3 plots the 99th percentile rather than the mean.

The correction works by emptying the tail. Comparing the last two column pairs: the corrected predictor moves 86.19% of values into categories 0–2, all of which cost four bits, and cuts categories 5–8 from 5.53% to 1.00% of values. It is not that individual residuals got a little smaller; it is that the expensive ones almost stopped occurring.

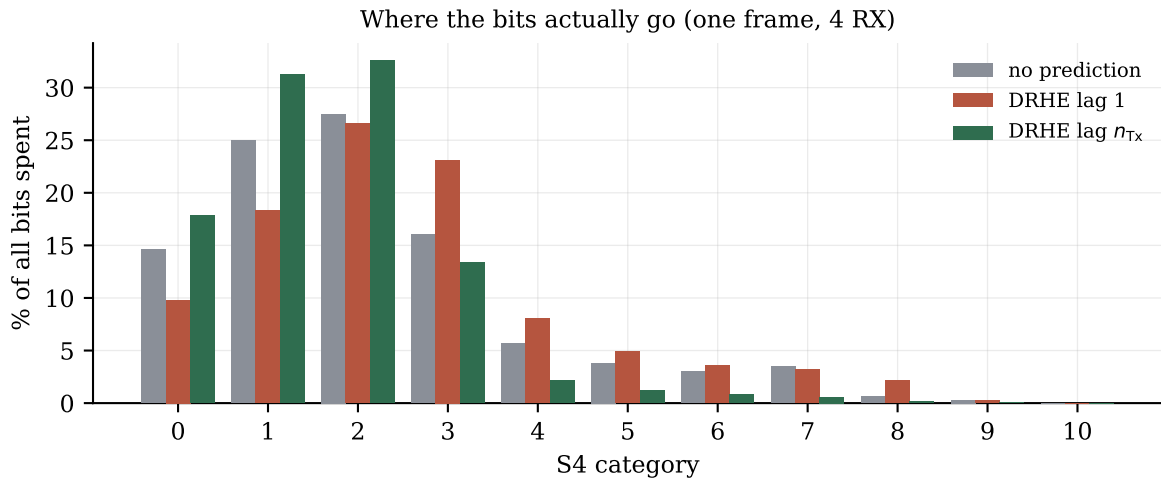


Figure 3.2: The “% bits” columns of Table 3.6 drawn. The corrected predictor concentrates spending into the three four-bit categories; the lag-1 predictor pushes it out into categories 3 through 8, where each value costs two to four times as much.

The lag-1 predictor loses to doing nothing in a specific way. Against the no-prediction column it *gains* slightly in categories 0–2 but loses heavily in 3, 4 and 8. It is redistributing values upward, not downward. A predictor that helps should hollow out the right-hand side of this table; this one fills it in.

3.1.11 How good is the coder itself?

It is worth separating the coder’s performance from the predictor’s, since the two are independent and only one of them turned out to be the problem.

The order-0 entropy of the raw FX16 values — the theoretical floor for any code that treats values independently, from Equation (2.9) — is 4.094 bits. The coder achieves 4.576 bits on the same data. That is an overhead of 0.48 bits, or 11.8 %, for a dictionary fixed at design time against a different dataset; on the five-frame average used in Chapter 6 the figure is 6.2 %.

That is a good result, and it has a clear implication: **there is very little left to win in the entropy coder.** Even a perfect, data-derived, per-frame-optimal code would improve the compression ratio by only a few per cent. All the remaining leverage is in the predictor — which is what makes a predictor that was contributing nothing such an expensive thing not to have noticed.

One caveat is worth recording for future work. The dictionary gives its two-bit codewords to categories 2, 3 and 4, which suits a distribution centred there. After the prediction-axis correction the distribution has moved *down*: categories 0, 1 and 2 now hold 86 % of values, and category 0 — exactly zero — is paying four bits for what is now the second most common symbol. Re-deriving the dictionary for the corrected residual distribution is an obvious and cheap improvement that this thesis does not make.

3.1.12 A representability hole

There is one value the code cannot represent, and it caused a real defect.

Category 15 spans $|v| \in [16384, 32767]$. Counting both signs that is exactly $2^{15} = 32,768$ values, which exactly fills the fifteen payload bits — the tightness that Table 3.2 noted as a virtue leaves no spare codepoint. But `int16` arithmetic can produce $v = -32768$. Applying Equation (3.5) gives

$$A(-32768) = -32768 + 32767 = -1, \quad (3.9)$$

which truncated to fifteen bits is 0x7FFF — bit-for-bit the same payload as $v = +32767$, with the same category and therefore the same codeword. The two are indistinguishable on the wire, and the decoder returns +32767.

This is not hypothetical. Section 5.4.1 reports it firing in a synthetic edge case with a reconstruction error of 65,535, which is precisely the distance from -32768 to $+32767$ — a diagnostic number that pointed straight back to this paragraph.

3.2 DRHE: Doppler Redundancy Hybrid Encoding

3.2.1 Why predict in polar coordinates

Section 2.1 established what a range-bin sequence looks like across slow time for a target at constant velocity: the *magnitude* is nearly constant, and the *phase* advances by a constant increment. In Cartesian (I, Q) coordinates that is a point rotating on a circle — both components oscillate, and a simple difference between consecutive samples is large. In polar coordinates the same sequence is two nearly-trivial signals: a constant and a ramp.

DRHE therefore converts to polar, predicts magnitude and phase separately with filters matched to those two shapes, converts the prediction back to Cartesian, and subtracts.

3.2.2 The two predictors

For ramp m and range bin n , with $a[m, n]$ the magnitude and $\theta[m, n]$ the phase, the predictions are

$$\hat{a}[m, n] = \alpha \hat{a}[m-1, n] + (1 - \alpha) a[m-1, n], \quad (3.10)$$

$$\hat{\theta}[m, n] = \beta \hat{\theta}[m-1, n] + (2 - \beta) \theta[m-1, n] - \theta[m-2, n], \quad (3.11)$$

with $\alpha = 0.6$ and $\beta = 0.4$ [11, Table 4.4], and $\hat{\theta}$ wrapped into $[-\pi, \pi]$.

Equation (3.10) is a first-order infinite-impulse-response smoother: it blends the previous prediction with the previous observation. For a constant magnitude it converges to that constant, and the α term gives it some immunity to noise on any single ramp.

Equation (3.11) is more interesting. Setting $\beta = 0$ reduces it to $\hat{\theta}[m] = 2\theta[m-1] - \theta[m-2]$, which is a *double integrator*: it extrapolates the last two phases along a straight line. For a phase advancing by a constant $\Delta\theta$ per ramp — exactly Equation (2.5) — that prediction is perfect. The β term mixes in the previous prediction to damp the response to noise.

The prediction is then converted back and subtracted, with int16 wraparound $\mathcal{W}(\cdot)$:

$$d_{\Re} = \mathcal{W}(x_{\Re} - \text{round}(\hat{a} \cos \hat{\theta})), \quad (3.12)$$

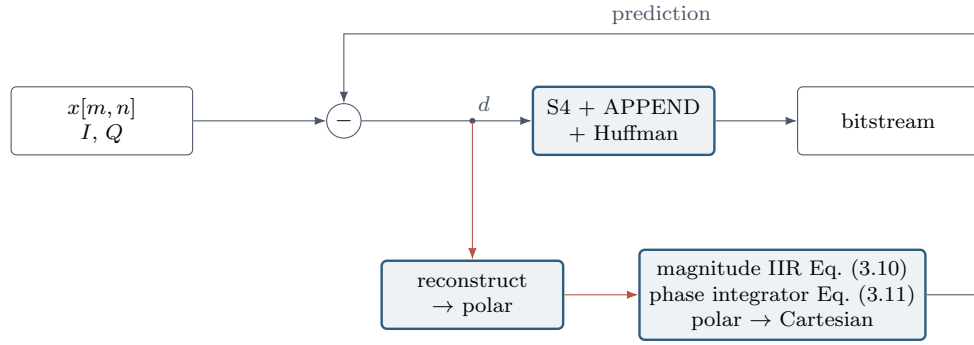
$$d_{\Im} = \mathcal{W}(x_{\Im} - \text{round}(\hat{a} \sin \hat{\theta})), \quad (3.13)$$

and $d_{\Re}$, $d_{\Im}$ are handed to the entropy coder of Section 3.1.

3.2.3 The streaming property

DRHE has a property that matters more than its compression ratio: it is *causal and streaming*. Each output depends only on the current sample and on state carried from ramp $m-1$. Concretely, the implementation holds five floating-point values per (receive channel, range bin) — the magnitude prediction, the previous magnitude, the phase prediction, and the previous two phases — and nothing else.

It therefore needs no frame buffer, and it emits its first compressed bits as soon as the first sample arrives. For a compressor sitting between an FFT and a memory bus, that is exactly the right shape. Section 6.7 returns to this when weighing DRHE against LPC.



closed loop: the encoder's state is driven by what the *decoder* will reconstruct

Figure 3.3: DRHE as implemented. Prediction happens in the polar domain with two filters matched to the expected behaviour of a constant-velocity target; the prediction is converted back to Cartesian before subtraction. The red path is the closed-loop correction of Section 5.4.1: the encoder drives its state from what the decoder will reconstruct, not from the original sample.

3.2.4 The assumption

DRHE's correctness as a *compressor* — it is lossless regardless — is not in question, but its *effectiveness* rests on one assumption made implicitly by the indices in Equations (3.10) and (3.11): that ramp $m - 1$ is the slow-time predecessor of ramp m .

Section 2.2.3 showed that on a TDM-MIMO cascade it is not.

3.3 LPC: linear prediction with Yule–Walker coefficients

3.3.1 Autoregressive modelling

The second algorithm takes a more general approach. Rather than assuming a particular signal shape and hand-designing filters for it, it *fits* a model to the data.

A second-order autoregressive (AR) model says each sample is approximately a linear combination of the previous two:

$$\hat{x}[m] = a_1 x[m-1] + a_2 x[m-2]. \quad (3.14)$$

Given a sequence, the coefficients that minimise the mean-square prediction error satisfy the Yule–Walker equations, which for order two are

$$\begin{bmatrix} R_0 & R_1 \\ R_1 & R_0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}, \quad (3.15)$$

where $R_k = \sum_m x[m]x[m-k]$ is the autocorrelation at lag k . Solving directly,

$$a_1 = \frac{R_1(R_0 - R_2)}{R_0^2 - R_1^2}, \quad a_2 = \frac{R_0 R_2 - R_1^2}{R_0^2 - R_1^2}. \quad (3.16)$$

3.3.2 Stability

An AR model whose poles lie outside the unit circle is unstable: prediction errors grow rather than decay. For order two the stability triangle gives the conditions

$$|a_2| < 1 \quad \text{and} \quad |a_1| < 1 - a_2. \quad (3.17)$$

If either fails, or if the denominator $R_0^2 - R_1^2$ vanishes, the implementation sets $a_1 = a_2 = 0$, which degrades gracefully to “predict zero” — i.e. code the raw value. On the real data used here the test never fires: 100.00% of coefficient sets pass.

Table 3.7: The two algorithms compared on assumptions and structure, before any measurement.

	DRHE	LPC
Prediction domain	polar (magnitude, phase)	Cartesian, I and Q independently
Model	fixed IIR filters, coefficients α, β chosen a priori	second-order AR, coefficients fitted per frame
What it assumes	constant magnitude, constant phase increment	the sequence is well described by an order-2 AR process
Causal?	yes — streaming, first output immediate	no — needs the whole frame before any output
State	5 values per (channel, bin)	3.1 Mibit frame buffer
Side information	none	32,768 bits/frame (1.04 %)
Per-sample arithmetic	4 floating-point transcendental (<code>sqrt</code> , <code>atan2</code> , <code>sin</code> , <code>cos</code>)	2 fixed-point multiply-accumulates
Entropy back end	identical: S4 + APPEND + fixed Huffman	

3.3.3 Granularity and side information

The model is fitted per (*range bin, receive channel, real/imaginary part*), across all ramps of the frame. That granularity is a deliberate compromise: fitting one model to the whole frame would ignore that different range bins contain targets at different velocities, while fitting per (bin, ramp) would have no data to fit to.

The coefficients are side information: the decoder cannot reproduce them, so they must be transmitted. Two 16-bit values per (bin, channel, part) is

$$128 \times 4 \times 2 \times 2 \times 16 = 32,768 \text{ bits} \quad (3.18)$$

per frame, which is 1.04 % of the uncompressed frame. That is a small but real tax, and Section 6.5 shows a variant where it becomes the dominant cost and sinks the whole approach.

3.3.4 The cost: LPC is not causal

R_0 , R_1 and R_2 are sums over *all* ramps of the frame. The coefficients therefore cannot be known until the frame is complete, and the compressor cannot emit a single bit before then. Where DRHE holds five values per (channel, bin), LPC must buffer the entire frame: $4 \times 192 \times 128 \times 32 = 3.1$ Mibit.

This is the fundamental trade between the two algorithms, and Section 6.7 argues it matters more than the difference in compression ratio.

3.4 Side-by-side

Table 3.7 summarises. The expectation going in — stated explicitly here so that Chapter 6 can be checked against it — was that LPC would compress slightly better, because a fitted model should beat a fixed one, but would be the expensive design, because it contains a division and DRHE does not. Exactly one half of that expectation survived measurement.

Chapter 4

Methodology

4.1 The verification ladder

A compression algorithm that is *nearly* lossless is useless, so the method has to be able to prove exactness rather than estimate it. The approach taken is a ladder of four phases, each one validating the phase above it at a lower level of abstraction. The key property is that each rung has a *reference* it can be checked against, so a discrepancy always localises to one transition.

Phase 1 — MATLAB. The entire radar chain is modelled in double precision, and all eight compression schemes are run against it. This establishes the golden reference: the exact FX16 values, the exact number of bits each algorithm should produce, and the detection metrics.

Phase 2 — HLS C-simulation. Because the hardware design is written in C++, it can be compiled and executed as ordinary software. This phase asks one question: does the C++ produce *exactly* the bits that MATLAB says it should, and does the decompressor recover *exactly* the original samples? It runs in minutes over all 50 frames, which makes it the workhorse of the whole project.

Phase 3 — RTL co-simulation. The C++ is synthesised to Verilog and the same testbench is re-run with the generated RTL substituted for the software. This checks that synthesis preserved behaviour — pipelining, array partitioning and fixed-point inference can all change results if a design is subtly wrong — and returns a cycle-accurate latency figure that no software simulation can give.

Phase 4 — silicon. Out of scope here; it requires the physical board. Every hardware number in this thesis is therefore a synthesis or co-simulation *estimate*, which Section 7.4 restates as a limitation.

4.2 The reference signal chain

The MATLAB chain is described here in the order it executes, because several later observations depend on details of individual steps.

4.2.1 Loading

A frame is read from the ColoRadar binary and reshaped into a complex cube of (fast time $\times$ ramp $\times$ receiver), with the TDM flattening of Equation (2.7). The per-ramp mean is

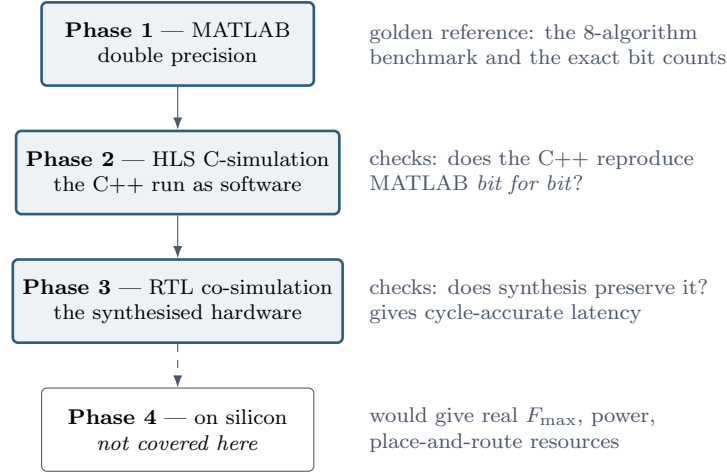


Figure 4.1: The verification ladder. Each phase is checked against the one above it, so a mismatch localises to a single transition rather than to “somewhere in the design”.

subtracted from each channel to remove the DC / transmitter-leakage term.

4.2.2 Windowing

A Hann window is applied along fast time to suppress spectral leakage, so that a strong nearby target does not bury a weak distant one in sidelobes.

4.2.3 Range FFT

An N_s -point FFT is taken along fast time and normalised by N_s . Only the positive half-spectrum is retained, giving $N = 128$ range bins. This halving is a deliberate choice, not a consequence of conjugate symmetry — the input here is complex I/Q , so the spectrum is *not* symmetric; the negative half corresponds to negative beat frequencies, which are not physically meaningful ranges.¹

4.2.4 FX16 quantisation

The complex FFT output is scaled by Equation (2.11) and rounded to `int16`. This is the last step before compression, and its output — 128 range bins $\times$ 192 ramps $\times$ 4 receive channels of complex `int16` — is exactly what both compressors consume.

Four receive channels rather than sixteen are exported to the hardware path, because four complex `int16` samples is 128 bits, the AXI4-Stream width. Section 7.4 notes the consequence for comparisons against the 16-channel MATLAB benchmark.

4.2.5 Detection and metrics

The reconstructed data is put through Doppler and angle processing and a threshold detector, and the resulting detections are compared against those from the double-precision reference. The threshold is the mean of the lowest 75% of sorted magnitudes plus a fixed 15 dB offset, with 8-connected local-maximum suppression.

¹A code comment in the reference implementation described this as a symmetry consequence. It is not, and the comment has been corrected.

4.3 Metrics, defined precisely

Several metric names are used loosely in this field. They are pinned down here.

4.3.1 Compression ratio

$$\text{CR} = \frac{B_{\text{in}}}{B_{\text{out}}}, \quad B_{\text{in}} = N_{\text{ramps}} \times N \times N_{\text{RX}} \times 32, \quad (4.1)$$

with B_{out} the compressed length *including* AXI packet padding. Including the padding matters: a design that flushed its bit buffer per ramp rather than per frame would waste up to 255 bits per ramp, costing about 2.9% of the ratio. Reporting the ratio without padding would hide that design choice.

4.3.2 Maximum absolute difference

$$\text{MaxDiff} = \max_{\text{all samples}} |x - \tilde{x}| \quad (4.2)$$

over every real and imaginary component. For a lossless coder the pass criterion is $\text{MaxDiff} = 0$. There is no tolerance.

4.3.3 Detection metrics, and one correction

Let D_{ref} be the detection set from the double-precision reference and D_{test} that from the reconstructed data. The two rates reported are

$$\text{FN}\% = \frac{|D_{\text{ref}} \setminus D_{\text{test}}|}{|D_{\text{ref}}|}, \quad \text{FP}\% = \frac{|D_{\text{test}} \setminus D_{\text{ref}}|}{|D_{\text{test}}|}. \quad (4.3)$$

The first is a false negative rate in the ordinary sense. The second is **not** a false positive rate, despite the label it carries in the literature and in this author's own earlier reporting. A false positive rate normalises by the number of true negatives — in a range–Doppler map, the number of empty cells. This quantity normalises by the number of *test detections*, which makes it a **false discovery rate**. It is a perfectly legitimate metric, and arguably the more useful one here, but it is not what its conventional name denotes. It is reported under its correct name throughout this thesis.

Two further properties bound how much weight the detection metrics can carry, and are stated rather than buried:

- **The comparison is cell-wise and exact.** A target detected one range–Doppler cell away from its reference position counts as both a false negative *and* a false discovery. There is no positional tolerance, which inflates both rates relative to a matched-detection criterion.
- **The 15 dB threshold is a free parameter.** It is not swept, and the detection rates are sensitive to it.

4.3.4 What the detection metrics actually measure here

This point is important enough to state on its own. FX16, DRHE and LPC are all exactly lossless *with respect to the FX16 codes*. Their reconstructed data is bit-identical, so their detection metrics are identical *by construction* — not by coincidence, and not as evidence that the compressors are equally good.

What the detection metrics measure in this thesis is therefore the cost of the **FX16 quantisation step**, which is the only lossy operation in the lossless chain. They are reported for that purpose, and to provide the contrast against the genuinely lossy BAQ.

Table 4.1: The synthetic edge-case suite. Each case has its own allowed reconstruction error — zero for all but the two that exercise the saturation described in Section 5.4.1.

Case	What it exercises	Allowed MaxDiff
all-zeros	the $S_4 = 0$ path and the best case	0
dc-constant	a perfectly predictable sequence	0
random-int16	the worst case: incompressible input	0
tone-192ramp	a periodic sequence the AR model should fit	0
tone-1ramp	a sequence too short to fit	0
max-amplitude	full-scale values, and 128-bit correlations	0
max-alternating	± 32767 alternating: the largest residuals	1
int16-min	-32768 specifically	1

4.3.5 Order-0 entropy

To judge the entropy coder rather than the whole compressor, Equation (2.9) is evaluated empirically on the residual distribution. The gap between the entropy and the achieved bits per value is the coder’s inefficiency; the entropy itself is the figure of merit for the *predictor*.

This is the measurement that exposed the central finding, and it is worth noting that it is not a standard part of a compression evaluation. Had it not been made, the prediction-axis error would have remained invisible behind a respectable-looking compression ratio.

4.4 Edge-case testing

Real data exercises the typical case well and the extreme case not at all. The ColoRadar residuals span roughly $[-514, +416]$, which is entirely inside category 10 of a 16-category code. Everything above that was, until deliberately tested, unexercised.

An eight-case synthetic testbench was therefore built (Table 4.1), generating its stimulus internally so that it runs identically in C-simulation and co-simulation.

Both defects reported in Section 5.4 were found by this suite and by nothing else. Neither is reachable with real radar data. The general lesson — that dataset-driven verification and adversarial synthetic verification find different classes of bug and neither substitutes for the other — is returned to in Section 7.5.

4.5 Toolchain and reproducibility

All MATLAB work used R2024b with the Signal Processing Toolbox. All hardware work used AMD Vitis HLS 2026.1 targeting `xcku5p-ffvb676-2-e` at a 100 MHz clock, with a licensed Vivado installation for synthesis and XSIM for co-simulation.

Two practical points are recorded because they cost time. First, Vitis refuses to create a project under a path containing a space, so the build tree must live somewhere like `D:\Thesis` rather than under a user directory whose name contains one. Second, the stimulus for the hardware testbenches is exported *from MATLAB as `int16`* rather than recomputed in C++. This eliminates any possibility of a floating-point FFT discrepancy between the two environments: the C++ testbench and the MATLAB reference operate on byte-identical input, so any difference in output is unambiguously a difference in the compressor.

Chapter 5

Implementation

5.1 Common infrastructure

Both compressors present the same interface: AXI4-Stream in at 128 bits per beat (four channels of 16-bit I and Q), AXI4-Stream out at 256 bits per beat of compressed bitstream, with the frame dimensions (N , N_{ramps} , N_{RX} , and for the corrected variants n_{TX}) on an AXI4-Lite control bundle.

Both share the bit-packing arrangement. Codewords accumulate into a 512-bit register, least-significant-bit first, and a 256-bit AXI word is flushed whenever at least 256 bits are available, with a final partial flush at end of frame. Flushing per frame rather than per ramp is what avoids the 2.9% padding penalty noted in Section 4.3.

The Huffman codewords are stored bit-reversed relative to the published dictionary, because the packer is LSB-first. The reversed code set remains prefix-free, so the decoder can match candidates from the low end of its buffer without ambiguity — at most one table entry can match.

5.2 DRHE architecture

5.2.1 State and parallelism

DRHE holds five `float` arrays of shape $[N_{\text{RX}}][N]$: the magnitude prediction, the previous magnitude, the phase prediction, and the previous two phases. The channel dimension is completely partitioned, so the four receive channels are four parallel copies of the datapath inside one pipelined iteration of the sample loop — the parallelism argument of Section 2.7 made concrete.

State is zeroed on the first ramp of each frame. This makes frames independent: a frame lost in transit cannot corrupt its successors, which is a desirable property for a streaming sensor and one that the 50-frame test confirms directly.

5.2.2 Why it is expensive

Per sample, per channel, DRHE evaluates `sqrt` and `atan2` to get polar coordinates, then `sin` and `cos` to get back. Four floating-point transcendental functions, four times over for four channels, on every one of 24,576 beats per frame.

Chapter 6 shows the bill: these functions account for the bulk of 115,534 LUTs, and the resulting combinational depth is why the design estimates 81.88 MHz rather than meeting its 100 MHz target. Replacing them with a fixed-point CORDIC would be the obvious optimisation and is left as future work.

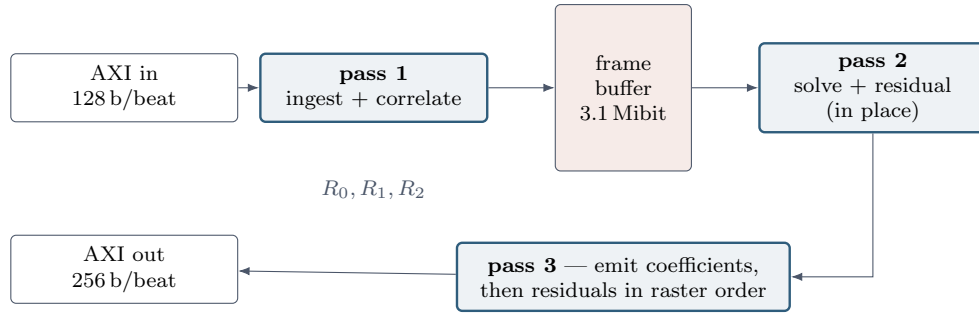


Figure 5.1: The LPC compressor. Because the Yule–Walker coefficients depend on the whole frame, the compressor must buffer it; the buffer is then rewritten in place with residuals so that only one memory is needed. Emitting residuals in the original raster order is what lets the *decompressor* avoid a frame buffer entirely.

5.2.3 The initiation-interval bottleneck

DRHE achieves $II = 2$ rather than the target $II = 1$, and the reason is instructive because it has nothing to do with arithmetic.

Each state array is partitioned on the channel dimension, so each channel’s $[N]$ history is one dual-port block RAM. Every iteration reads a value from it and writes the updated value back — two accesses, which two ports can just manage. But on the first ramp the iteration *also* writes a reset value. Three accesses against two ports forces the tool to serialise, and the loop settles at $II = 2$ for the whole frame on account of a condition that is true for one ramp in 192.

The fix is straightforward — hoist the reset into its own unpipelined loop before the main one, or partition cyclically on the bin dimension as well — and would roughly double throughput. It is identified rather than implemented here.

5.3 LPC architecture

5.3.1 Three passes

LPC cannot stream, so its structure is explicitly multi-pass (Figure 5.1):

1. **Ingest.** Read the frame into the buffer, accumulating R_0, R_1, R_2 per (channel, bin) as the samples arrive.
2. **Solve and residual.** For each (channel, bin), solve Equation (3.16), then walk the ramps again computing residuals and rewriting the buffer *in place*.
3. **Emit.** Write the coefficient table, then the residuals.

Rewriting the buffer in place matters: it means one 3.1 Mibit memory serves both purposes rather than two.

5.3.2 A useful asymmetry

Pass 3 emits the coefficient table first — the decoder needs all of it before it can reconstruct anything — and then the residuals *in the same raster order the samples arrived in*, rather than in the (channel, bin, ramp) order pass 2 produced them.

The payoff is that the decompressor needs no frame buffer at all. It holds the coefficient table and two reconstructed samples per (channel, bin) — the same *shape* of state DRHE uses — and writes each output sample as its residual arrives. The reordering cost is paid once, by the side that already had to buffer.

This is worth noting because it inverts the usual expectation. LPC’s non-causality is a compressor-side problem only; at the receiving SoC, where memory is comparatively plentiful but latency still matters, LPC decompression is as cheap as DRHE’s.

5.3.3 Fixed-point choices

Coefficients are quantised to `ap_fixed<16,2>` for a_1 (range $[-2, 2)$) and `ap_fixed<16,1>` for a_2 (range $[-1, 1)$), which follows from the stability conditions of Equation (3.17). Measured over 50 frames, this quantisation costs 0.0000 % of compression ratio relative to double-precision coefficients — the residuals are unchanged to the bit. The fitted models are evidently not sensitive at that level, which retrospectively justifies the 16-bit side-information budget.

The autocorrelations are accumulated as exact 64-bit integers rather than in floating point. This is deliberate: $R_0^2 - R_1^2$ is a difference of two nearly equal quantities whenever the data is strongly correlated — precisely the regime the predictor exists for — so computing it in floating point would lose most of its significant digits exactly when accuracy matters most. Forming all three 128-bit quantities exactly in integer arithmetic before any division avoids the cancellation entirely.

5.4 Two defects found and fixed

Both defects described here share a pattern worth naming: each was invisible on real data, each was found by a synthetic edge case, and each would have shipped.

5.4.1 Defect 1: the -32768 residual

Symptom. The max-alternating edge case — input alternating ± 32767 — produced $\text{MaxDiff} = 65,535$. That number is diagnostic: it is exactly the distance from -32768 to $+32767$, which points straight at the representability hole of Section 3.1.12. Instrumenting the encoder confirmed it:

```
MISMATCH v=-32768 s4=15 append=0x7FFF decoded=32767
```

The three possible fixes.

1. **Extend the code with a category 16.** Correct in general, but it changes the bitstream format and breaks compatibility with the reference design’s dictionary.
2. **Reject the input.** Legitimate for a closed system but unhelpful for a general IP core.
3. **Saturate -32768 to -32767 .** Costs at most one LSB on the affected sample and leaves the format untouched.

Option 3 was chosen. On real data the clamp never fires, since residuals span roughly $[-514, +416]$, so behaviour on the dataset is unchanged.

Why saturation alone is not enough. Saturation makes the reconstruction differ from the original by one LSB, and that is where Section 2.4.4 becomes concrete. The encoder originally derived its next prediction from the *original* sample, while the decoder necessarily derives it from the *reconstructed* one. These agree only while reconstruction is exact. The moment the clamp fires they diverge — and because Equations (3.10) and (3.11) are IIR filters, that divergence is then carried forward through every remaining ramp of the frame, growing rather than decaying.

The complete fix is therefore to close the loop. The encoder reconstructs exactly what the decoder will produce and drives its state from that:

```

int diff_re = wrap_int16(re - pred_re);
if (diff_re == -32768) diff_re = -32767;      // saturate
int recon_re = wrap_int16(diff_re + pred_re); // what the decoder will see
float curr_mag = hls::sqrt(recon_re*recon_re + recon_im*recon_im);
float curr_phase = hls::atan2(recon_im, recon_re); // state from recon

```

The cost. Where reconstruction is exact — which is everywhere on real data — the closed-loop and open-loop forms are identical by construction, so compression is unchanged. The cost falls entirely on synthesis: the reconstruction now sits on the dependency chain ahead of the polar conversion, so pipeline depth rises from 32 to 58 cycles and registers from 40,309 to 57,105. Initiation interval and $F_{\max}$ are unaffected.

That is a real price — Section 7.1 notes that it roughly doubles the IP-core latency — and it is the direct cost of bounding an error that cannot occur on this dataset but could occur on another. Option 1 above would avoid it, at the price of bitstream compatibility.

5.4.2 Defect 2: 128-bit to float conversion

Symptom. The `max-amplitude` edge case produced wrong coefficients. Tracing showed a determinant of 4.4×10^{20} converting to -1.2×10^{18} — not merely imprecise but the wrong sign.

Cause. Equation (3.16) needs products of 64-bit correlations, so the numerators and the determinant are computed as `ap_int<128>`. Converting such a value directly to `float` is not correct above 2^{64} in this toolchain: `(float)(1 << 70)` evaluates to zero.

Fix. Normalise all three operands by a common arithmetic right shift into 62 bits before the division, choosing the shift from the largest of the three:

$$k = \max(\text{msb}|D|, \text{msb}|N_1|, \text{msb}|N_2|) - 61, \quad (5.1)$$

clamped at zero. Because the same shift is applied to numerator and denominator, the quotient is mathematically unchanged; only the intermediate representation is brought into a range the conversion handles.

Why it was invisible. On real ColoRadar data the correlations reach only 44 bits, comfortably inside the range where the conversion is correct. The defect is unreachable with this dataset and would have been unreachable with any similarly scaled one. It is, in other words, precisely the class of bug that a dataset-driven test plan cannot find.

5.5 The TDM-aware variants

Chapter 6 motivates these; this section describes what they are, since they are implementation rather than result.

5.5.1 DRHE

The change is confined to state indexing. The five state arrays gain a transmitter dimension, indexed by $m \bmod n_{\text{Tx}}$, and the reset condition widens from $m = 0$ to $m < n_{\text{Tx}}$, because each transmitter's history begins on its own first ramp rather than on the frame's. The prediction equations, the coefficients, the residual computation, the entropy coder and the bitstream format are all untouched.

The state grows by a factor n_{Tx} in principle. In practice the baseline was dimensioned for up to 1,024 range bins and the corrected version for the 128 actually used, so the realised growth is 1.5 $\times$: from 40 to 80 block RAMs.

Crucially, the *schedule* does not change. Indexing the state by transmitter changes which address is read, not when, so the initiation interval, the pipeline depth and the frame latency are all identical. The correction is therefore close to free.

5.5.2 LPC, and a ring buffer trick

LPC needs more work, because both the autocorrelation and the residual pass now need $x[m - n_{Tx}]$ and $x[m - 2n_{Tx}]$ rather than $x[m - 1]$ and $x[m - 2]$.

A naive shift register of depth $2n_{Tx} = 24$ would shift 24 entries every iteration. Instead, observe that a *ring* of exactly $2n_{Tx}$ entries gives both taps for free. At ramp m , let $k = m \bmod 2n_{Tx}$. Slot k still holds the value written at ramp $m - 2n_{Tx}$, because it is about to be overwritten and has not been touched since; and slot $(k + n_{Tx}) \bmod 2n_{Tx}$ holds the value from ramp $m - n_{Tx}$, since that is the most recent ramp congruent to $k + n_{Tx}$. So:

read $x[m - 2n_{Tx}]$	from slot k
read $x[m - n_{Tx}]$	from slot $(k + n_{Tx}) \bmod 2n_{Tx}$
write $x[m]$	to slot k

Nothing is shifted; the cost is $2n_{Tx}$ registers and two multiplexers.

5.5.3 A restructure that pays for itself

Accumulating the autocorrelations during ingest is no longer convenient at the deeper lags, so they were moved into the per-bin pass instead, computed from the frame buffer through the ring above. This deleted six 64-bit accumulator arrays and four history arrays outright.

The consequences, quantified in Chapter 6, are mixed and interesting: block RAM usage *falls* by 25%, and the ingest loop — previously stuck at $\Pi = 3$ on accumulator port pressure — now meets $\Pi = 1$, as does every other loop in the design. Against that, the design now makes two short passes over each (channel, bin) instead of one long one, and Section 6.6 shows that the pipeline drain this incurs costs considerably more than a naive cycle model predicts.

5.5.4 Both baselines retained

The original DRHE and LPC designs were kept alongside the corrected ones rather than replaced. This is deliberate: it allows a like-for-like comparison at every level, and it means the results in Chapter 6 are differences measured on the same machine with the same toolchain on the same day, not comparisons against remembered numbers.

Chapter 6

Results

This chapter reports what was measured. It is arranged so that the story unfolds in the order it actually did: first the benchmark and the verification, which all passed and all looked satisfactory; then the audit that asked what the numbers meant and found they did not mean what they appeared to; then the correction and its consequences.

6.1 Phase 1: the MATLAB benchmark

6.1.1 Eight algorithms on real data

Table 6.1 gives the 50-frame average over 9,934 reference detections, using all 16 receive channels. The reference against which everything is compared is the double-precision path, so FP16 — half-precision floating point — registers as effectively lossless.

Three things are worth drawing out.

The lossless coders share FX16’s detection metrics exactly. FX16, DRHE and LPC report identical FN%, FP%, noise floor and SNR. As Section 4.3.3 explained, this is not a coincidence but a tautology: all three reconstruct the FX16 codes bit-exactly, so the detector sees identical data. What the table is really showing in those rows is the cost of *quantisation*, which is 3.77 % of detections missed and 1.11 % of reported detections spurious.

The lossy comparison is instructive. BAQ reaches a nominally better ratio — 3.76 against 3.30 — but misses 10.44 % of detections instead of 3.77 %, and 9.69 % of what it does report is spurious against 1.11 %. That is nearly three times the miss rate and nearly nine times the false discovery rate, for a 14 % improvement in ratio. In a system that brakes the car, that is not a trade anyone should take. This is the empirical form of the argument made on principle in Section 1.3.

The predictive coders reach the target. DRHE’s 3.30 matches the 3.2 reported for the baseline design [11], which at the time was taken as confirmation that the implementation was correct. Section 6.4 revisits that interpretation.

6.2 Phase 2: bit-exactness

6.2.1 The headline

Over 50 frames — 157,286,400 complex samples — both HLS implementations reconstruct every sample of every frame exactly: MaxDiff = 0, mean squared error = 0, signal-to-noise ratio infinite. Compression ratios are 3.30987 for DRHE and 3.38497 for LPC.

Table 6.1: Phase 1: eight compression schemes, 50-frame average, 16 receive channels, 9,934 reference detections. FP% is a false discovery rate (Section 4.3.3). NF is the noise floor.

Algorithm	CR	FN%	FP%	NF (dBFS)	SNR (dB)
FP16	1.00	0.00	0.00	−105.638	35.504
DRHEfp16	1.04	0.05	0.01	−105.637	35.505
FX16	1.00	3.77	1.11	−105.374	35.359
DRHE	3.30	3.77	1.11	−105.374	35.359
BAQ	3.76	10.44	9.69	−105.080	35.081
RDVLE	1.12	24.01	4.62	−102.275	33.165
RDVLE_UDRE	2.56	28.73	4.44	−101.782	33.004
LPC	3.37	3.77	1.11	−105.374	35.359

Table 6.2: LPC, mean compression ratio over 50 frames, computed four ways. The padding model reproduces the hardware result on every frame, so the HLS bit counts are bit-exact with MATLAB.

Quantity	Value
MATLAB, double-precision coefficients	3.38541
MATLAB, 16-bit quantised coefficients	3.38541
(quantisation cost)	0.0000 %
MATLAB, quantised + 256-bit AXI padding	3.38497
Vitis HLS, measured	3.38497
Frames where the padding model matches	50 / 50
Worst-frame difference	0.00091 (tolerance ± 0.01)

6.2.2 How closely the hardware matches MATLAB

Agreement with the MATLAB reference is an order of magnitude tighter than the ± 0.01 criterion set in advance.

For LPC the agreement is exact. A MATLAB model that charges the same 256-bit AXI padding reproduces the Vitis bit count on 50 of 50 frames to within 4.9×10^{-6} — which means the entire discrepancy between the two is packet padding and the underlying bit counts are identical.

For DRHE the hardware bit count exceeds the MATLAB count by between 15 and 252 bits per frame, mean 134. That range sits strictly inside one 256-bit packet, which is exactly what a per-frame flush should produce, so again the residual difference is padding rather than disagreement.

6.2.3 Edge cases

All eight synthetic cases of Table 4.1 pass for both coders after the fixes of Section 5.4. Two results are worth recording because they were initially misread as failures:

- **random-int16 compresses to 0.60** — that is, it *expands* the data by 67%. This is correct behaviour, not a bug. Incompressible input has no structure to remove, and every residual lands in a high category costing a 12-bit codeword plus 15 APPEND bits. Every lossless coder expands incompressible input; it is a theorem that some input must.
- **tone-192ramp reaches only 1.10 under LPC**. The synthetic tone’s period is about 209 ramps, longer than the 192-ramp frame, so the autocorrelation estimate is poor and the fitted coefficients are $a_1 = 1.0377$, $a_2 = -0.0484$ against the ideal $a_1 = 2 \cos \omega = 1.9991$, $a_2 = -1$. MATLAB’s solver produces the same coefficients, and its stability test does not fire either, so this is a property of the second-order Yule–Walker estimator on a

Table 6.3: Post-synthesis resource and timing estimates, `xcku5p` at a 100 MHz target. Percentages are of the device, which has 216,960 LUT, 433,920 FF, 960 BRAM_18K and 1,824 DSP. The reference column is Kiem’s ZU3EG design [11], which is a device roughly three times smaller, so only absolute counts are comparable.

Resource	DRHE	LPC	Reference [11]
LUT	115,534 (53%)	80,528 (37%)	45,892
FF	57,105 (13%)	17,393 (4%)	9,243
BRAM_18K	40 (4%)	256 (26%)	10
DSP	132 (7%)	168 (9%)	44
Est. $F_{\max}$	81.88 MHz	85.43 MHz	100 MHz
Frame latency	61,253 cyc	197,277 cyc	—
Worst loop II	2	3	1
Frame buffer	none	3.1 Mibit	—

truncated period, not an implementation fault.

6.3 Phase 3: synthesis and co-simulation

6.3.1 Resources

Table 6.3 gives post-synthesis estimates for both compressors on the `xcku5p`.

6.3.2 Neither design meets timing

Both estimate below 100 MHz: 81.88 MHz for DRHE and 85.43 MHz for LPC. The critical path in DRHE is the chain of floating-point transcendentals described in Section 5.2. The remedy — pipelining the float datapath, or replacing the polar conversion with a fixed-point CORDIC — is identified but not implemented.

6.3.3 The expectation that did not survive

Section 3 recorded the prior expectation that LPC would be the expensive design, because it contains a division and DRHE does not.

It is not. LPC uses 30 % fewer LUTs and 70 % fewer flip-flops. The reason is that per-sample cost dominates: LPC’s inner loop is two fixed-point multiply–accumulates, while DRHE’s is four floating-point transcendentals. The division runs 1,024 times per frame — once per (bin, channel, part) — and contributes 512 cycles out of 197,277, or 0.26 %. It is not the bottleneck by any measure.

What LPC costs instead is *memory and latency*. Its frame buffer accounts for 240 of its 256 block RAMs, and the frame must be complete before the first output bit exists. The real trade between the two algorithms is therefore **logic against memory and latency**, which was not the trade either was expected to present.

6.3.4 Where LPC’s cycles go

The cycle count decomposes exactly, which is a useful check that nothing unexpected is happening:

The model is accurate to 0.08 %. Two inefficiencies are visible and neither is algorithmic: the ingest loop sits at $II = 3$ on accumulator port pressure where 24,576 cycles should do, and the residual pass handles one channel at a time where ingest and emit handle all four. Fixing both would bring LPC to roughly 74,000 cycles, within 20 % of DRHE, without touching the algorithm.

Stage	Iterations	II	Cycles
Ingest	24,576	3	73,728
Solve + residual	98,304	1	98,304
Emit coefficients	512	1	512
Emit residuals	24,576	1	24,576
Model total			197,120
Measured (co-simulation)			197,277

Table 6.4: What happens when the predictor is removed. “No prediction” codes the FX16 values directly through the same S4 + Huffman back end. Five frames, four receive channels. LPC includes its coefficient overhead.

Configuration	bits/value	CR	vs. baseline
No prediction (entropy coder only)	4.586	3.489	—
DRHE, lag 1 (as implemented)	4.821	3.319	0.951×
LPC, lags 1 & 2 (as implemented)	4.710	3.397	0.974×

At this point every test had passed, both designs were verified bit-exact on real data, and the compression ratios matched the literature. The obvious conclusion was that the work was done.

6.4 The audit: what the compression ratio actually measures

6.4.1 The question

A compression ratio of 3.3 on radar data invites a particular reading: that the predictor is removing about 70 % of the information, and that the Doppler redundancy the algorithm was designed around is really there and really being exploited.

That reading deserves a test, and the test is simple. Both compressors use the same entropy coder. So: code the FX16 output through that coder with *no prediction at all*, and see what ratio survives. Whatever survives is attributable to the entropy coder and to the statistics of the data — not to prediction.

6.4.2 The result

Table 6.4 is the pivot of this thesis. Coding the data with *no predictor at all* compresses **better** than either predictor. DRHE costs 4.9 % more bits than doing nothing; LPC costs 2.6 % more.

Order-0 entropy confirms the direction independently of the coder:

Quantity	Value
Order-0 entropy of raw FX16 codes	4.094 bits
Order-0 entropy of DRHE residuals	4.538 bits
Achieved bits/value (no prediction)	4.586 bits
Huffman overhead above entropy	0.283 bits (6.2 %)

The predictor is *increasing* the entropy of the data it is supposed to be decorrelating. That is not a marginal inefficiency; it is a predictor that is actively wrong.

Note also that the entropy coder itself is doing fine: 4.586 bits achieved against 4.094 bits of entropy is an overhead of 6.2 %, which for a fixed dictionary designed for a different dataset is a good result. The problem is not in the second stage.

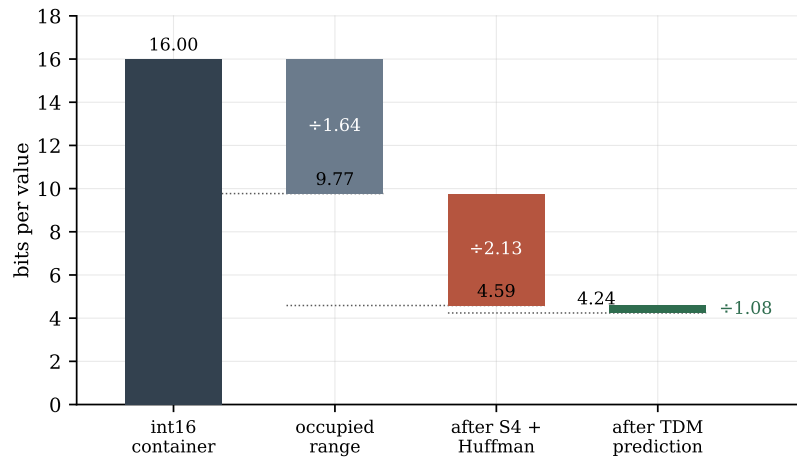


Figure 6.1: Where the compression ratio comes from. Of the 16 bits in the `int16` container, 6.2 are structurally unused because the capture sits far below full scale; the entropy coder recovers a further factor of 2.13 from spectral sparsity; and prediction *on the corrected axis* contributes the remaining 1.08. Prediction on the original ramp axis contributes nothing at all — it costs 0.24 bits per value.

6.4.3 Where the ratio really comes from

If not prediction, then what? Two places, neither of which is Doppler redundancy (Figure 6.1).

Unused container headroom: a factor of 1.63. Equation (2.11) calibrates the FX16 scale factor to a theoretical full-scale ADC sinusoid. The ColoRadar capture sits far below full scale: the peak raw ADC sample is 914 out of a possible 32,767. After windowing, transform and scaling, the peak `int16` FFT code is 436. Only $\log_2(436) + 1 = 9.8$ of the 16 container bits ever carry information; the remaining 6.2 are zero in every single sample. An entropy coder removes them whether or not any prediction happens, and that alone is a factor of $16/9.8 = 1.63$.

Spectral sparsity: a factor of 2.14. The remaining $9.8 \rightarrow 4.59$ bits is the genuine statistical gain, and it is a real property of radar data: a range-FFT frame is sparse. Most bins contain noise near zero and a few contain targets, so the S_4 categories concentrate heavily at the low end. This is what the entropy coder is for and it does it well.

Prediction: nothing, as implemented. The third factor should have been the predictor's contribution. As implemented it was negative.

Consequence for reporting. The compression ratios in Table 6.1 — and, by the same argument, those reported elsewhere in this field — are substantially a property of the *sensor's gain staging* rather than of the algorithm. A capture with 6 dB more gain would compress measurably less on the same algorithm. A no-prediction baseline should be reported alongside any such figure, and is not standard practice.

6.5 The prediction axis

6.5.1 Diagnosis

The inversion in Table 6.4 is not a property of the predictors — they are both sound algorithms — but of the *axis* they were applied along. Section 2.2.3 set this up: both were predicting from

Table 6.5: Prediction axis study. Five frames, four receive channels, 128 bins, 192 ramps. Prediction gain is relative to no prediction at all; a gain below 1.0 means the predictor is making things worse.

Configuration	bits/value	CR	gain
No prediction	4.586	3.489	1.000
DRHE, lag 1 (<i>as implemented</i>)	4.821	3.319	0.951
DRHE, lag n_{Tx}	4.240	3.774	1.082
LPC, lags 1, 2 (<i>as implemented</i>)	4.710	3.397	0.974
LPC, per-transmitter blocks, $N = 16$	6.093	2.626	0.753
LPC, lags $n_{Tx}, 2n_{Tx}$	4.269	3.748	1.074

ramp $m - 1$, which by Equation (2.7) is a different physical transmitter eleven times in twelve.

The mechanism is now concrete. Two ramps from different transmitters differ by the target’s array steering phase, Equation (2.6), which is large and varies with target bearing. DRHE’s phase model, Equation (3.11), is a double integrator that predicts perfectly when the phase increment is constant; a per-ramp transmitter change makes that increment alternate on a period-12 pattern, which no double integrator can track. LPC’s autocorrelations at lags 1 and 2 likewise measure the *transmitter switching pattern* rather than target Doppler, and the fitted coefficients describe the former.

6.5.2 Testing the hypothesis

Six configurations were evaluated over five frames (Table 6.5).

Predicting on the true slow-time axis converts DRHE’s 4.9 % loss into an 8.2 % gain, and LPC’s 2.6 % loss into a 7.4 % gain. The hypothesis is confirmed.

6.5.3 The variant that does not work, and why

Row five is worth dwelling on because it is the obvious alternative and it fails badly.

Splitting the frame into twelve independent 16-ramp per-transmitter blocks and fitting separate coefficients to each *does* put the predictor on the right axis. But it multiplies the coefficient side information by twelve — from 1.04 % to 12.5 % of the uncompressed frame — and it shortens each autocorrelation estimate from 192 samples to 16, which makes the fitted coefficients much noisier. The net result, CR 2.626, is far worse than doing nothing at all.

Keeping *one* coefficient pair per (bin, channel, part) and simply moving the lags to n_{Tx} and $2n_{Tx}$ costs no extra side information whatsoever, which is why the last row works and the fifth does not. The lesson generalises: on a TDM-MIMO radar, the right change is to the *lag*, not to the *partitioning*.

6.5.4 What the residuals look like

The bits-per-value figures are a summary; the underlying change in the residual distribution is more revealing.

Figures 6.2 and 6.3 show what happened. Lag-1 prediction *widens* the residual distribution: the 99th percentile of $|\text{residual}|$ goes from 80.6 with no prediction to 102.4 with DRHE. Predicting at lag n_{Tx} collapses it to 15.8, and the corrected LPC reaches 8.2 — roughly a tenfold reduction in tail magnitude against the original design.

The category histogram tells the same story in the currency the coder actually charges in. Lag-1 DRHE moves 23.0 % of residuals into category 3 and 6.9 % into category 4, which cost 5 and 6 bits; the corrected version puts 84.6 % of residuals into categories 0–2, all of which cost 4 bits or fewer.

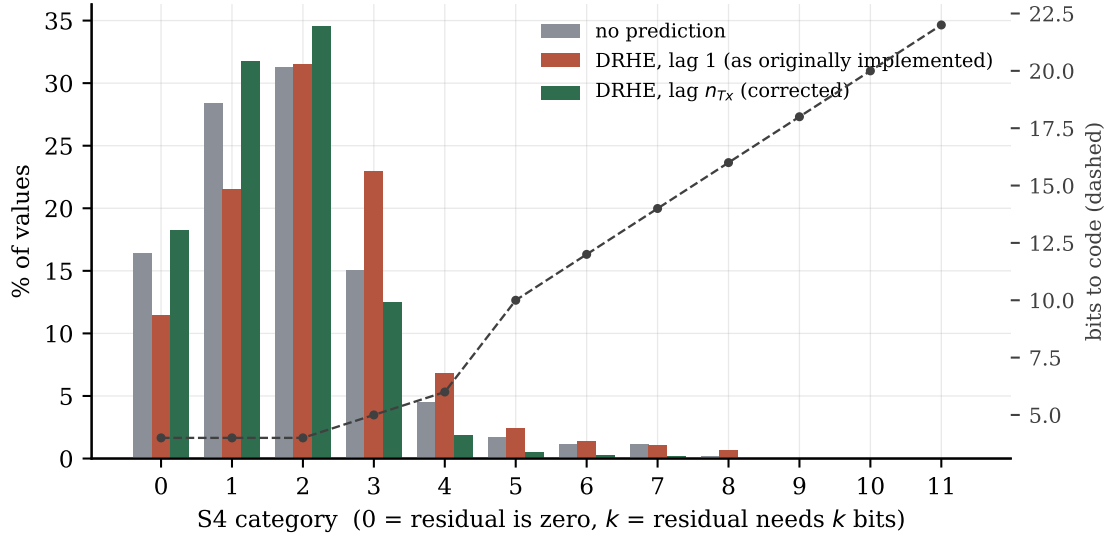


Figure 6.2: Distribution of residuals across S_4 categories, with the cost of each category (Equation (3.7)) on the right axis. Lag-1 DRHE (red) pushes mass *up* the category scale relative to not predicting at all (grey) — into categories 3 and 4 which cost 5 and 6 bits. Lag- n_{Tx} DRHE (green) pulls it back down into categories 0–2, which cost 4 bits or fewer. The steepness of the cost curve above category 4 is what makes this shift so consequential.

Table 6.6: The corrected designs against their baselines. Vitis HLS C-simulation, all 50 frames, four receive channels. Every configuration reconstructs every sample exactly.

Design	CR	MaxDiff	Δ CR
DRHE, lag 1	3.30987	0	—
DRHE, lag n_{Tx}	3.75082	0	+13.3%
LPC, lags 1, 2	3.38497	0	—
LPC, lags $n_{Tx}, 2n_{Tx}$	3.69681	0	+9.2%

6.6 The corrected implementations

The corrections described in Section 5.5 were implemented in HLS and put through the identical verification ladder as the baselines: 50-frame C-simulation, synthesis, and RTL co-simulation.

6.6.1 Compression, measured in hardware

The measured 3.75082 for the corrected DRHE agrees with the 3.774 predicted by the five-frame MATLAB study of Table 6.5, to within the frame-to-frame spread. That agreement between an independent MATLAB prediction and a hardware measurement is the cross-check that the correction is real rather than an artefact of either environment.

Figure 6.5 shows the per-frame detail. The improvement is uniform: there is no frame on which the corrected design does worse, and the shape of the variation across the sequence is preserved, which is what one would expect if the correction improves prediction rather than merely shifting a constant.

6.6.2 What the correction costs

DRHE: close to free. Forty additional block RAMs — 4% of the device — buys 13.3% more compression. LUTs rise by 204 (0.18%), flip-flops by 158, DSPs not at all, and $F_{\max}$

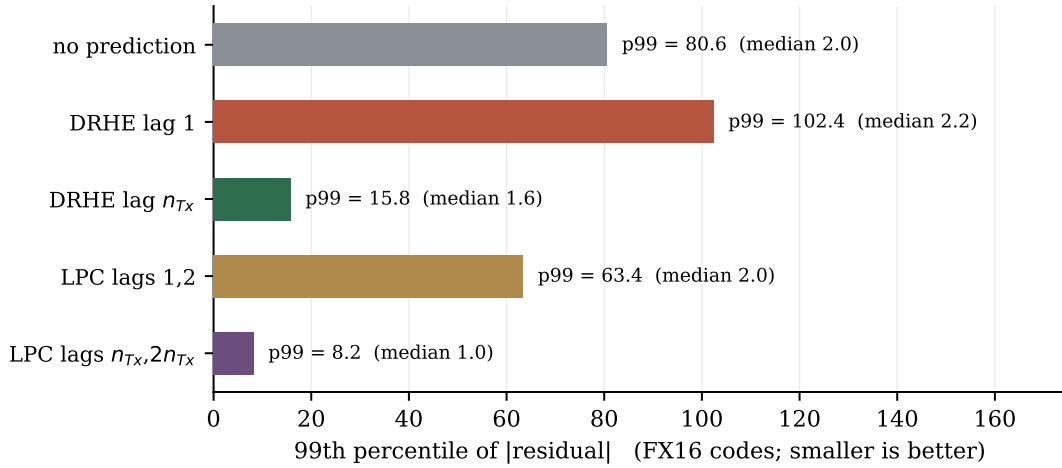


Figure 6.3: The 99th percentile of residual magnitude. This is the clearest single view of the effect: lag-1 DRHE produces a heavier tail than no prediction at all (102.4 against 80.6), while lag- n_{Tx} DRHE cuts it to 15.8 — a factor of 6.5 better than the design it replaces. LPC at lags $n_{Tx}, 2n_{Tx}$ reaches 8.2.

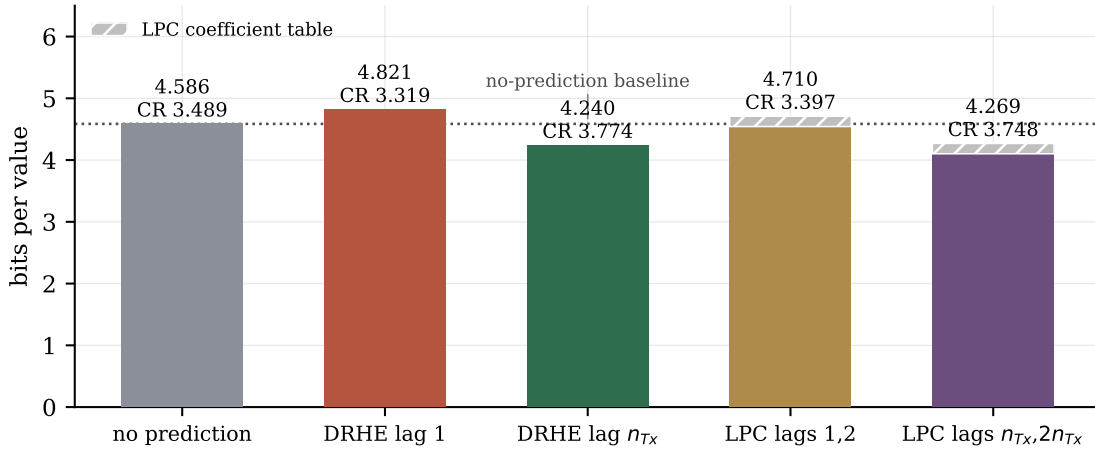


Figure 6.4: Bits per value for all five configurations, with the LPC coefficient table shown separately as it is charged whatever the residuals do. The dotted line is the no-prediction baseline: any bar above it is a predictor that is doing harm.

is unchanged to the reported precision. RTL co-simulation measures the frame latency at exactly 61,253 cycles for *both* the baseline and the corrected design.

That last point is not a coincidence but a structural consequence: indexing the predictor state by transmitter changes *which* address is read, not *when*, so the schedule the tool generates is identical. The entire 13.3% gain is bought with block RAM and nothing else.

LPC: a genuine trade. The LPC picture is more interesting, because the restructure that the deeper lags required turned out to fix an unrelated problem. Moving the autocorrelation out of the ingest pass deleted six 64-bit accumulator arrays and four history arrays, so block RAM usage *falls* by 25%, from 256 to 192. More importantly, the ingest loop — previously stuck at $II = 3$ on accumulator port pressure — now meets $II = 1$, and so does every other loop in the design. The synthesis report states that all loop constraints are satisfied, which the baseline never achieved.

Against that: 7% more LUTs and 33% more flip-flops for the $2n_{Tx}$ -entry rings, and — the real cost — considerably more latency.

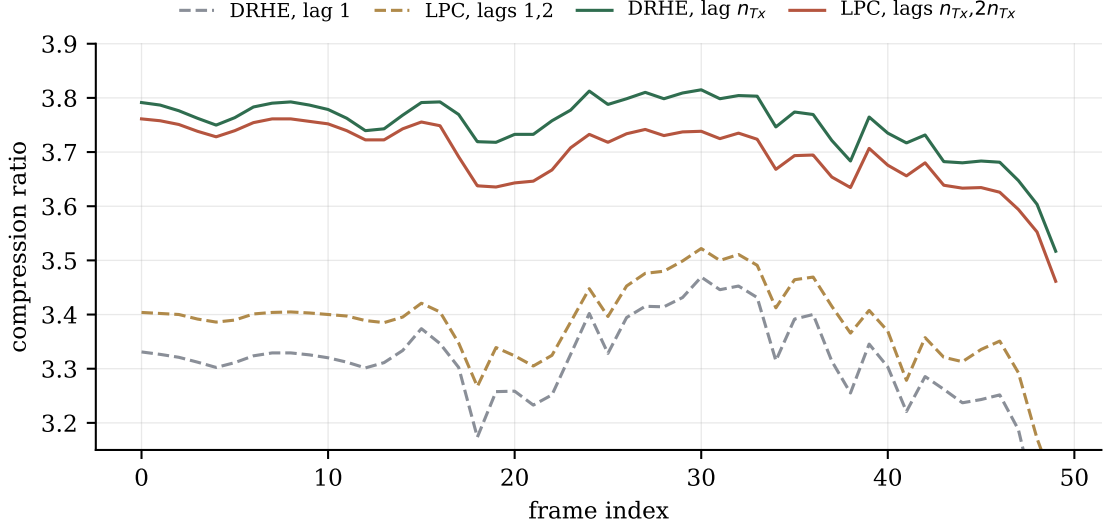


Figure 6.5: Per-frame compression ratio across the 50-frame sequence, measured in C-simulation. The corrected designs (solid) sit above their baselines on *every* frame, not merely on average, and the frame-to-frame variation — which tracks scene content — is preserved. All four configurations reconstruct every sample exactly.

Table 6.7: Cost of the correction, `xcku5p` at a 100 MHz target. Resources and $F_{\max}$ are from synthesis; frame latency is measured in RTL co-simulation on a single frame.

Resource	DRHE		LPC	
	lag 1	lag n_{Tx}	lags 1, 2	lags $n_{Tx}, 2n_{Tx}$
LUT	115,534	115,736	80,528	86,298
FF	57,105	57,263	17,393	23,154
BRAM_18K	40	80	256	192
DSP	132	132	168	156
$F_{\max}$ (MHz)	81.88	81.88	85.43	85.43
Worst loop II	2	2	3	1
Frame latency (cyc)	61,253	61,253	197,277	325,166

6.6.3 A caution about cycle models

The latency result deserves its own discussion, because it is a methodological lesson as much as a number.

For the baseline designs the obvious cycle model — trip count times initiation interval, summed over loops — is accurate to 0.08 %, as Section 6.3 showed. Applying the same model to the corrected LPC predicts 246,272 cycles. Co-simulation measures 325,166. The model is wrong by 32 %.

The missing term is **pipeline drain**. Moving the autocorrelation into the per-bin pass replaced one long pipelined loop with *two short loops invoked 512 times each*. A pipelined loop pays its depth once on entry before it reaches steady state — 71 cycles for the correlation loop and 56 for the residual loop — so 512 invocations contribute roughly

$$512 \times (71 + 56) \approx 65,000 \text{ cycles} \quad (6.1)$$

of pure drain that no II-based model accounts for. That closes most of the gap.

So the corrected LPC is *slower* than the design it replaces: 325,166 cycles against 197,277, a factor of 1.65. This is a real cost and is stated as such rather than buried. It is, however, structural rather than algorithmic: merging the two per-bin loops, or processing several bins

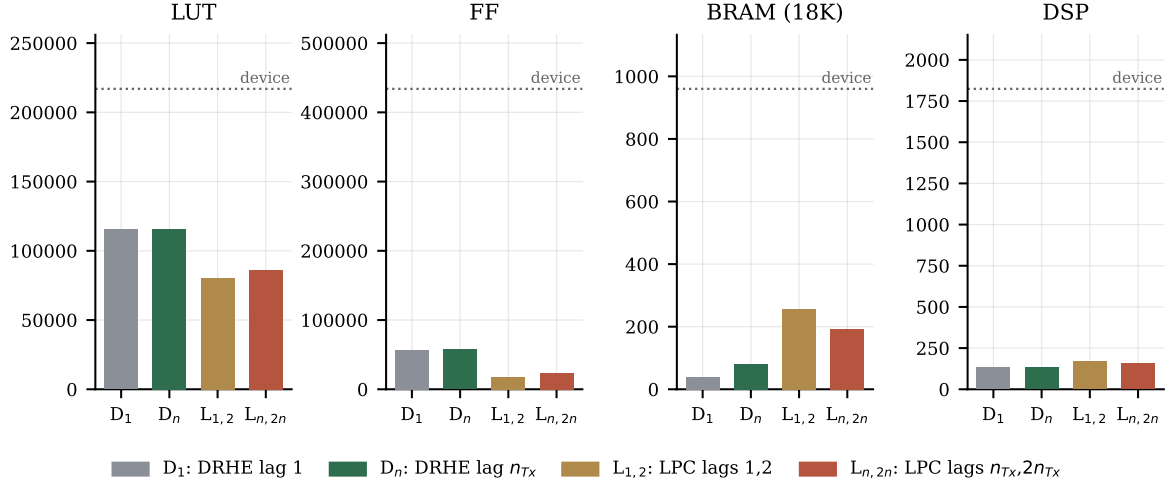


Figure 6.6: Resource usage for the four designs against the xcku5p capacity (dotted). DRHE trades block RAM for logic; LPC trades logic for block RAM. The TDM correction moves DRHE’s BRAM from 40 to 80 and, by removing the ingest accumulators, moves LPC’s *down* from 256 to 192.

Table 6.8: The two corrected designs compared. Bold marks the better of the two on each row.

	DRHE, lag n_{Tx}	LPC, lags $n_{Tx}, 2n_{Tx}$
Compression ratio	3.75082	3.69681
Lossless	yes	yes
LUT	115,736	86,298
FF	57,263	23,154
BRAM_18K	80	192
DSP	132	156
$F_{\max}$	81.88 MHz	85.43 MHz
Frame latency	61,253 cyc	325,166 cyc
Worst loop II	2	1
Compressor frame buffer	none	3.1 Mibit
First output available	immediately	after a full frame
Side information	none	1.04 %

per loop invocation so the drain amortises, would recover most of it. DRHE pays no such price at all.

6.6.4 Co-simulation

Both corrected designs pass RTL co-simulation with $\text{MaxDiff} = 0$, so losslessness survives synthesis as well as it did in software. The co-simulation is run on a single-frame stimulus, as is standard for a cycle-accurate run: 50 frames of RTL simulation with floating-point IP takes many hours of wall-clock time for no additional information, since the latency per frame is deterministic.

6.7 DRHE against LPC, after the correction

After the correction the ranking on compression ratio reverses: DRHE is ahead, 3.751 against 3.697, where before it was behind. That is because the correction helps DRHE more (13.3%) than LPC (9.2%), which makes sense — DRHE’s phase model is *specifically* a constant-

Doppler model, so putting it on the Doppler axis recovers exactly what it was designed to find, whereas LPC’s general AR model was always going to fit *something*.

On logic LPC is clearly cheaper. On memory and latency DRHE is clearly better, and by a wide margin: no frame buffer at all, first output immediately, and a fifth of the frame latency.

For the application that motivates this thesis — a compression IP sitting between the range FFT and the memory bus in a streaming sensor front end — latency and streaming behaviour outrank a 1.5 % difference in ratio. **DRHE is the better choice**, and after the correction it happens to win on ratio too. Where a frame of latency is acceptable and block RAM is plentiful, LPC’s logic saving is real and worth having.

6.8 Summary of results

1. Both compressors are exactly lossless over 157 million real samples, verified at three levels of abstraction.
2. As originally implemented, both predictors performed *worse than no prediction*: CR 3.319 and 3.397 against 3.489 for the entropy coder alone.
3. The cause is the transmitter-interleaved ramp axis of a TDM-MIMO cascade. Predicting at lag n_{Tx} raises measured compression to 3.751 (DRHE) and 3.697 (LPC), still exactly lossless.
4. For DRHE the correction costs 40 block RAMs and nothing else — logic, F_{max} and measured frame latency are unchanged.
5. For LPC the correction is a trade: better ratio, 25 % less block RAM, $\text{II} = 1$ throughout, against $1.65\times$ the frame latency.
6. The ratio decomposes as $1.63\times$ from unused container headroom, $2.14\times$ from spectral sparsity, and $1.08\times$ from prediction once it is applied correctly.

Chapter 7

Discussion

7.1 Comparison with the baseline design

Comparing against Kiem’s implementation [11] is informative but not like-for-like: different device, different radar, different dataset, different detection pipeline. Two corrections are needed before any comparison is meaningful at all, and both were initially got wrong in this project.

7.1.1 Device size

An early draft of this work compared utilisation percentages directly, on the assumption that the `xcku5p` and the ZU3EG had comparable capacity. They do not. The `xcku5p` has 216,960 LUTs against the ZU3EG’s 70,560 — roughly three times larger — and correspondingly more flip-flops, block RAM and DSP slices.

Percentage utilisations are therefore meaningless as a comparison, and only absolute counts mean anything. In absolute terms this DRHE implementation uses $2.5\times$ the LUTs of the reference. That is a worse result than the percentage comparison suggested, and it is the correct one.

7.1.2 Throughput units

The reference reports 11.92 Gbit/s at 100 MHz. Taken as decimal gigabits that is

$$\frac{11.92 \times 10^9}{100 \times 10^6} = 119.2 \text{ bits per cycle} \quad (7.1)$$

on a 128-bit input — an awkward number with no architectural interpretation. Taken as *gibibits*, however,

$$\frac{11.92 \times 2^{30}}{100 \times 10^6} = 127.99 \text{ bits per cycle}, \quad (7.2)$$

which is 128 to within rounding: exactly the full width of the AXI4-Stream input, at $\text{II} = 1$. The reference figure is therefore in binary units, and its decimal equivalent is 12.80 Gbit/s.

This matters because any comparison against 11.92 understates the gap by 7.4 %. Table 7.1 gives the corrected comparison.

7.1.3 The gap is structural, not mysterious

Being $2.5\times$ slower than the reference sounds alarming until it is decomposed, at which point it becomes a list of known, fixable things.

The reference achieves $\text{II} = 1$. This DRHE achieves $\text{II} = 2$ on the block-RAM port conflict described in Section 5.2, and loses a further 20 % to per-ramp pipeline drain. Fixing the port

Table 7.1: Throughput comparison at 100 MHz. The reference column restates the published figure in decimal units. Both designs here are the baselines; the corrected DRHE is identical in throughput.

	DRHE	LPC	Reference [11]
Effective II	2	3	1
bits per cycle	51.4	15.9	128.0
Throughput	5.14 Gbit/s	1.60 Gbit/s	12.80 Gbit/s
IP pipeline latency	580 ns	—	230 ns

conflict — by partitioning the state arrays cyclically on the bin dimension, or by hoisting the first-ramp reset out of the pipelined loop — would bring this design to 11 to 12 Gbit/s, in line with the reference.

LPC’s inefficiencies are similarly addressable, and Section 6.6 showed that the TDM restructure has already resolved its $II = 3$ ingest.

The pipeline latency of 580 ns against the reference’s 230 ns is a different matter: it is 320 ns inherent plus the cost of the closed-loop correction of Section 5.4.1, which deepened the pipeline from 32 to 58 cycles. That is the price of bounding an error the reference design does not bound. Adopting a category-16 extension instead would recover it, at the cost of bitstream compatibility.

7.2 What the compression ratio means elsewhere

Section 6.4 showed that 1.63 of the 3.489 no-prediction ratio is unused container headroom. This has a consequence that should be stated plainly: **these ratios do not transfer to a differently-scaled sensor.**

A capture whose ADC output filled the `int16` container would have no structurally-zero top bits for the entropy coder to remove. Its compression ratio, on identical algorithms, would be closer to 2.1 — the sparsity factor alone — before prediction. That is still a useful ratio, and still enough to make the Gigabit Ethernet substitution of Section 1.2 work. But it is not 3.5.

Conversely, and more usefully, the headroom represents an opportunity rather than only a caveat. If the FX16 scale factor were made adaptive per frame — computed from the actual peak rather than from a theoretical full-scale sinusoid — the same 16 bits would carry roughly 6 more bits of genuine dynamic range. That directly addresses the quantisation-error problem that motivated this thesis’s first objective, since weak returns currently being crushed into the rounding noise of Equation (2.11) would survive. The reported compression ratio would fall, but the data being compressed would be better.

7.3 Does the correction generalise?

The prediction-axis result is specific in its details and general in its principle.

It is *specific* in that $n_{Tx} = 12$, and in that the fix assumes a fixed TDM rotation. A radar that changes its MIMO schedule between frames would need the lag re-loaded; one using Doppler-division or code-division multiplexing instead of TDM has no interleaved ramp axis and does not have this problem at all — though it has others.

It is *general* in that the underlying mistake is a category error that any slow-time predictive coder can make: treating an index that is *physically* a time axis as though it were *semantically* a time axis. The ramp index genuinely is time-ordered. It is simply not the case that adjacent entries are comparable, because the thing being multiplexed onto that axis changes between them. Any compression scheme inherited from a single-transmitter context and applied to a

multiplexed one is exposed to the same error, and the check — measure the entropy coder alone, and see whether the predictor beats it — costs almost nothing.

7.4 Threats to validity

These bound how far the results generalise and are stated explicitly rather than left implicit.

- **The compression ratios are capture-specific.** As above: most of the no-prediction ratio comes from an under-filled container, which is a property of this recording’s gain staging.
- **The detection metrics do not measure compression.** All three lossless schemes share identical detection metrics with FX16 by construction. What the metrics measure is the cost of the FX16 quantisation step.
- **FP% is a false discovery rate.** It normalises by test detections, not by true negatives. It is a legitimate metric but not the one its conventional name denotes, and it is reported under its correct name here.
- **Detection matching has no positional tolerance.** A target displaced by one range–Doppler cell is scored as both a miss and a false discovery, which inflates both rates relative to a matched-detection criterion.
- **The detection threshold is an unswept free parameter.** The mean of the lowest 75 % of magnitudes plus 15 dB is not varied, and the metrics are sensitive to it.
- **Channel count differs between phases.** The MATLAB benchmark of Table 6.1 uses all 16 receive channels; the hardware path uses 4, the AXI4-Stream width. Compression ratios between the two are therefore compared through a like-for-like MATLAB helper rather than directly.
- **All hardware numbers are pre-implementation estimates.** $F_{\max}$, resource counts and latency come from HLS synthesis and RTL co-simulation, not from place-and-route or silicon. Phase 4 remains outstanding, and post-route $F_{\max}$ is typically lower than the synthesis estimate.
- **Co-simulation covers one frame.** The 50-frame result is from C-simulation, which exercises identical logic but not identical timing. Nothing in either design is frame-count dependent, but this is an assumption rather than a measurement.
- **Two source citations required correction.** The BAQ baseline is properly attributed to Kwok and Johnson [22] — IEEE TGRS vol. 27, no. 4, pp. 375–383, 1989, on *synthetic* aperture radar. An earlier version of this work carried an author name, title and page range that do not correspond to any such paper. The ColoRadar citation is likewise vol. 41, no. 4, pp. 351–360 (2022) [12], not vol. 40 (2021). Both are corrected here.
- **The literature description of DRHE does not match the implementation.** A common summary of DRHE describes differencing *raw ADC* samples across chirps and coding the deltas with Golomb–Rice. The algorithm implemented and evaluated here — following [11] — operates on the *range-FFT output*, predicts in the polar domain with two IIR filters, and codes with a fixed Huffman dictionary over magnitude categories. The two make different assumptions about where the redundancy lives, and should not be conflated.

7.5 Lessons about verification

Three methodological observations seem worth recording, since they generalise beyond this project.

Dataset testing and adversarial testing find different bugs. Fifty frames of real data — 157 million samples — found neither defect in Section 5.4. An eight-case synthetic suite found both in minutes. The real data exercises the typical case exhaustively and the extreme case not at all: ColoRadar residuals occupy category 10 of a 16-category code, so a third of the coder was simply never executed. Conversely, the synthetic suite would never have found the prediction-axis error, because synthetic data has no transmitter structure to get wrong. Neither substitutes for the other.

A passing test suite is not a correctness argument. Everything passed. The compressors were bit-exact at three levels, the ratios matched the literature, the resources fitted the device. The prediction-axis error was invisible to every test in the plan, because no test asked whether the predictor was *helping* — only whether the system was lossless and whether the ratio was in the expected range. Adding a single measurement, the no-prediction baseline, exposed it immediately.

Cycle models are a good tool until they silently stop being one. Trip count times initiation interval predicted the baseline LPC latency to 0.08 %, which built entirely unwarranted confidence in it. Applied to a restructured design with many short loop invocations it was wrong by 32 %, because it omits pipeline drain — a term that is negligible in one topology and dominant in another. The lesson is not that the model is bad, but that a model validated on one design shape carries no guarantee on another.

Chapter 8

Conclusions and Future Work

8.1 What was done

This thesis set out to extend the radar compression framework established in [11] by validating it on real data, broadening the algorithm comparison, and implementing the complete compression and decompression pipeline. All three were done.

Two lossless compressors for radar range-FFT data — Doppler Redundancy Hybrid Encoding and a second-order linear predictive coder with Yule-Walker coefficients — were specified, modelled in MATLAB, implemented in C++ for Vitis HLS, synthesised for an AMD Kintex UltraScale+ device, and verified bit-exactly against a double-precision reference over 50 frames of real cascaded MIMO radar data at three levels of abstraction. Both reconstruct every one of 157,286,400 samples exactly, in software simulation and in RTL co-simulation alike. Matching decompressors were implemented for both, which is what made that verification possible.

8.2 What was found

8.2.1 The prediction axis

The principal finding is a correction rather than a confirmation. Applied the way the literature describes — predicting each ramp from its immediate predecessor — *neither predictor was contributing anything*. Coding the quantised FFT output through the identical entropy coder with no prediction whatsoever reaches a compression ratio of 3.489, against 3.319 for DRHE and 3.397 for LPC. Both were making the data measurably harder to compress, and the order-0 entropy of the residuals confirmed it independently: 4.538 bits for DRHE residuals against 4.094 bits for the raw codes.

The cause is that a twelve-transmitter time-division MIMO cascade interleaves its transmitters along the ramp axis, so “the previous ramp” belongs to a different physical antenna eleven times out of twelve — separated in space by the array baseline rather than in time by a pulse interval. Predicting instead from the previous ramp of the *same* transmitter puts both algorithms back on the genuine slow-time axis and recovers the gain they were designed to deliver. Measured in hardware simulation over the full dataset, compression rises from 3.310 to 3.751 for DRHE and from 3.385 to 3.697 for LPC, in both cases with every sample still reconstructed exactly.

For DRHE the correction costs 40 additional block RAMs — 4% of the device — with logic, clock frequency and measured frame latency all unchanged. For LPC it is a genuine trade: a better ratio, 25% less block RAM and $II = 1$ on every loop, against $1.65\times$ the frame latency.

8.2.2 What a compression ratio actually measures

The audit that exposed the above also decomposed the ratio into its genuine sources. Of the no-prediction factor of 3.489, roughly 1.63 comes from an under-filled fixed-point container — the ColoRadar capture peaks at 436 of a possible 32,767, so 6.2 of every 16 bits are structurally zero — and 2.14 from the genuine spectral sparsity of a range-FFT frame. Prediction, correctly applied, adds a further 1.08.

The practical recommendation that follows is simple and, as far as the author can establish, not current practice: **report a no-prediction baseline alongside any radar compression ratio**. Without it, a ratio conflates algorithm performance with sensor gain staging, and — as this thesis demonstrates — can conceal a predictor that is doing active harm.

8.2.3 Two defects, and two corrections to the record

Two implementation defects were found and fixed, neither reachable with real data and both caught by an eight-case synthetic suite: a residual value of -32768 that the entropy code cannot represent and which aliased silently onto $+32767$, resolved by saturation combined with closed-loop prediction so that encoder and decoder cannot diverge; and a 128-bit to single-precision conversion in the Yule-Walker solver that returned zero above 2^{64} , resolved by normalising all operands through a common arithmetic shift.

Separately, a throughput figure widely quoted for this class of design was found to be expressed in gibibits rather than gigabits: 11.92 Gibit/s is 12.80 Gbit/s decimal, which is exactly 128 bits per cycle at 100 MHz — a full-rate AXI4-Stream at $II = 1$. A detection metric conventionally labelled a false positive rate was identified as a false discovery rate, and two source citations were corrected.

8.2.4 Which algorithm

After the correction, DRHE is the better choice for the application that motivates this work. It compresses slightly better (3.751 against 3.697), it needs no frame buffer at all where LPC needs 3.1 Mibit, it emits its first bits immediately where LPC must wait a full frame, and its frame latency is a fifth of LPC's. LPC's advantage is logic — 30 % fewer LUTs and 60 % fewer flip-flops — which is real but which, for a streaming sensor-edge IP, does not outweigh a frame of latency.

The expectation recorded in Section 3 — that LPC would be the expensive design because it contains a division — did not survive measurement. The division runs 1,024 times per frame and contributes 0.26 % of the cycles. The expensive operations in this problem are `atan2` and the sine/cosine pair, not division.

8.3 Future work

8.3.1 Immediate

1. **Fix the DRHE initiation interval.** The $II = 2$ is a block-RAM port conflict caused by a reset that executes on one ramp in 192 but constrains the schedule for all of them. Hoisting the reset into its own loop, or partitioning the state cyclically on the bin dimension, should reach $II = 1$ and roughly double throughput to 11–12 Gbit/s.
2. **Amortise the LPC pipeline drain.** Merging the two per-bin loops, or processing several bins per invocation, should recover most of the 65,000 cycles of drain identified in Section 6.6.3.
3. **Replace the polar conversion with a fixed-point CORDIC.** Four floating-point transcendentals per sample dominate DRHE's LUT count and its critical path. A

CORDIC implementation should address both the resource usage and the $81.88 \text{ MHz } F_{\max}$.

4. **Complete Phase 4.** Place-and-route and on-silicon measurement to replace the estimates in Table 6.3, including power, which no simulation here reports.

8.3.2 Longer term

1. **Make the FX16 scale factor adaptive.** This is the highest-value single change the decomposition suggests. Using 9.8 of 16 bits wastes both dynamic range and honest reporting; a per-frame adaptive scale would let weak returns survive quantisation — directly addressing the quantisation-error problem that motivated this project’s first objective — and would make the reported ratio reflect algorithm performance rather than capture gain.
2. **Extend the correction to other multiplexing schemes.** Doppler-division and code-division MIMO do not produce an interleaved ramp axis, but they distribute the slow-time structure differently, and whether a slow-time predictor helps on them is an open question that the no-prediction baseline test would answer cheaply.
3. **Revisit the entropy coder for the corrected residuals.** The fixed Huffman dictionary was designed for a different residual distribution. After the prediction-axis correction, 84.6 % of residuals fall in categories 0–2 — a much more concentrated distribution than the dictionary assumes. Re-deriving the dictionary for this distribution, or adding a category-16 extension to close the representability hole of Section 3.1.12, would both be worth measuring.
4. **Evaluate the corrected algorithms against the neural and spectral approaches** of Section 2.8, which were out of scope here but define the upper end of achievable ratios.

8.4 Closing remark

The most useful thing this project produced was not either compressor. It was the habit of asking what a number means rather than whether it matches expectation. Every test passed, every ratio agreed with the literature, and the work appeared finished — at which point a single additional measurement, the cost of coding the data with no predictor at all, showed that the central mechanism of both algorithms had never been working. That measurement takes about twenty lines of MATLAB. It should be standard.

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Appendix A

The Huffman Dictionary

Table A.1 gives the complete fixed dictionary used by both compressors. The published codes [11, Table A.1] are most-significant-bit first; the implementation here packs least-significant-bit first, so the stored codewords are bit-reversed. Both forms are listed, and all sixteen entries were verified to agree.

The reversed code set remains prefix-free, which is what allows the decoder to match candidates from the low end of its bit buffer with no ambiguity.

Table A.1: The fixed Huffman dictionary. “Cost” is the total bits to code one residual in that category, $\ell + S_4$, from Equation (3.7).

S_4	$ v $ range	ℓ	code (MSB-first)	stored (LSB-first)	cost
0	$v = 0$	4	1010	0x0005	4
1	1	3	100	0x0001	4
2	2–3	2	01	0x0002	4
3	4–7	2	00	0x0000	5
4	8–15	2	11	0x0003	6
5	16–31	5	10110	0x000D	10
6	32–63	6	101110	0x001D	12
7	64–127	7	1011110	0x003D	14
8	128–255	8	10111110	0x007D	16
9	256–511	9	101111110	0x00FD	18
10	512–1023	10	1011111110	0x01FD	20
11	1024–2047	11	10111111110	0x03FD	22
12	2048–4095	14	10111111111111	0x3FFD	26
13	4096–8191	14	10111111111110	0x1FFD	27
14	8192–16383	13	1011111111110	0x0FFD	27
15	16384–32767	12	101111111110	0x07FD	27

Note the non-monotonicity at categories 12–15: the longest codewords are assigned to categories 12 and 13 rather than to 15. This is a property of the published dictionary and is reproduced faithfully. It has no effect on this dataset, where no residual exceeds category 10.

Appendix B

Bitstream Layouts

B.1 DRHE

DRHE’s bitstream is a single continuous sequence of coded residuals, in the raster order the samples arrive:

$$\underbrace{\text{enc}(d_{\mathfrak{R}}) \text{ enc}(d_{\mathfrak{S}})}_{\text{ramp 0, bin 0, ch 0}} \underbrace{\text{enc}(d_{\mathfrak{R}}) \text{ enc}(d_{\mathfrak{S}})}_{\text{ramp 0, bin 0, ch 1}} \cdots$$

where $\text{enc}(v)$ is the Huffman codeword for $S_4(v)$ followed by $S_4(v)$ APPEND bits. There is no header: the frame dimensions arrive over the AXI4-Lite control interface. The final AXI word is zero-padded.

B.2 LPC

LPC’s bitstream has two sections:

$$\underbrace{[a_1^{\mathfrak{R}} \ a_2^{\mathfrak{R}} \ a_1^{\mathfrak{S}} \ a_2^{\mathfrak{S}}]_{\text{ch,bin}}}_{\text{coefficient table, 4}\times\text{16 bits each}} \underbrace{\text{enc}(d_{\mathfrak{R}}) \text{ enc}(d_{\mathfrak{S}}) \cdots}_{\text{residuals, raster order}}$$

The coefficient table comes first because the decoder needs all of it before it can reconstruct anything. The residuals follow in the original raster order — not in the order the compressor computed them — which is what allows the decompressor to operate without a frame buffer (Section 5.3).

Appendix C

Reproducing the Results

C.1 Source files

File	Purpose
MATLAB reference	
loadColoRadarFrame.m	reads a frame, applies the TDM flattening
preprocessing.m	DC removal, bit shift, Hann window
rangeFFT.m	range FFT, positive half retained
compress_fx16.m	FX16 quantisation, Eq. (2.11)
export_frames_for_vitis.m	writes the HLS testbench stimulus
Audit scripts	
audit_decomp.m	the no-prediction baseline and entropy figures
audit_order.m	ramp-ordering demonstration
audit_predaxis.m	the six-configuration prediction-axis study
thesis_figdata.m	S_4 histograms and residual quantiles
HLS designs	
drhe_common.h	shared types, constants, Huffman table
drhe_compress.cpp	DRHE compressor (baseline)
drhe_decompress.cpp	DRHE decompressor
drhe_tdm_*.cpp/h	DRHE, corrected to lag n_{Tx}
lpc_common.h	LPC types and the Yule-Walker solver
lpc_compress.cpp	LPC compressor (baseline)
lpc_decompress.cpp	LPC decompressor
lpc_tdm_*.cpp/h	LPC, corrected to lags $n_{Tx}, 2n_{Tx}$
*_tb.cpp	50-frame testbenches
*_tb_edge.cpp	the eight-case synthetic suite

C.2 Build commands

C-simulation, synthesis and co-simulation respectively, for the corrected DRHE:

```
vitis-run --mode hls --csim --config test_tdm.cfg --work_dir hls_tdm
v++ -c --mode hls --config test_tdm.cfg --work_dir hls_tdm
vitis-run --mode hls --cosim --config test_tdm.cfg --work_dir hls_tdm
```

Two practical notes. Synthesis must be run before co-simulation, since co-simulation substitutes the generated RTL for the top function. And the project path must not contain a space, which Vitis rejects.

C.3 Co-simulation stimulus

Co-simulation is run against a single-frame export, produced by setting `DRHE_MAX_FRAMES=1` before running `export_frames_for_vitis.m`. Fifty frames of RTL simulation with floating-point IP takes many hours and returns no information the single-frame run does not, since the per-frame latency is deterministic. The 50-frame file should be restored afterwards for C-simulation.